

CERTIFIED COMPUTATION OF COMPONENTWISE SATURATION-HEIGHT BOUNDS IN THE CYCLOTOMIC $\mathbb{Z}_2$ - AND $\mathbb{Z}_3$ -TOWERS

HIROKI FUKUI

ABSTRACT. Let $B_n = \mathbb{Q}(2 \cos(2\pi/2^{n+2}))$ be the n -th layer of the cyclotomic $\mathbb{Z}_2$ -extension of $\mathbb{Q}$, h_n its class number and $k_n = h_n/h_{n-1}$. We present a criterion for $\ell \nmid k_n$ together with the certified computation that makes it effective: for every layer $n \geq 2$ and every odd prime ℓ , $\ell^{\text{ord}_{2^n}(\ell)} > C_n \Rightarrow \ell \nmid k_n$, with $C_n = 2(4/\pi)^{2^{n-2}} \Gamma(2+2^{n-2}) \prod_{\chi} |L(1, \chi)| / \log(2+\sqrt{5})^{2^{n-1}}$, the product over the even characters of conductor 2^{n+2} . The constant is evaluated by rigorous ball arithmetic and stored as exact dyadic outward endpoints; at $n = 7$ the criterion gives $\ell \nmid h_7/h_6$ for $\ell \equiv 63, 65, 127 \pmod{128}$, $\ell > 1.32 \times 10^{15}$, and for $\ell \equiv 1 \pmod{128}$, $\ell > 1.73 \times 10^{30}$. For the 1000 primes $\ell \equiv 65 \pmod{128}$ with $10^9 < \ell \leq 1001287361$, which Kashio and Yoshizaki excluded conditionally on their Conjecture 2.2, the exclusion is made unconditional by finite certificates checked by a read-only verifier whose soundness is proved relative to the same explicit literature inputs. The criterion is an instance of an abstract saturation–height bound (Theorem SH), which also gives, from an explicit bound for $|L(1, \chi)|$, a computation-free criterion at every layer, and applies in the cyclotomic $\mathbb{Z}_3$ -tower, where the covolume of the Horie-unit log lattice is computed exactly by a character product and is positive at every layer; at the layer of degree 81 the resulting layer-fixed bound lies below the uniform bound of Morisawa and Okazaki in both classes $\ell \equiv \pm 1 \pmod{81}$. Designated finite computations and discrete cores are checked by the Lean 4 kernel or by frozen certificate verifiers, the complete verification package is replayed by continuous integration from a clean checkout, and every external literature input is isolated as an explicit hypothesis.

CONTENTS

1.	The computational problem	2
1.1.	Weber’s problem and its known controls	3
1.2.	Statement of results	4
1.3.	Shape of the criteria	7
1.4.	The four surviving classes at $n = 7$	7

2020 *Mathematics Subject Classification*. Primary 11Y40, 11R18; Secondary 11R29, 11H06, 68V20.

Key words and phrases. Weber’s class number problem, cyclotomic $\mathbb{Z}_2$ -extension, relative units, saturation, Blichfeldt’s theorem, machine-checked certificates, Lean 4.

2. The componentwise criterion	9
2.1. Old and new components of the class group	9
2.2. The exponent ring, the log map and the components	9
2.3. Saturation	10
2.4. The height floor	11
2.5. The lattice of a component	12
2.6. Proof of Theorem A	12
3. Certificate soundness	13
3.1. Pipeline: untrusted generator, read-only verifier	13
3.2. Soundness	15
4. Exact evaluation of the constants	15
4.1. Proof guide for Proposition D	15
4.2. Numerical values	15
4.3. Algorithms and cost	16
5. The certified computation for the thousand primes	19
5.1. The family	19
5.2. Failure classes	21
5.3. Limits of the method	21
5.4. Resources	21
6. The cyclotomic $\mathbb{Z}_3$ -tower	22
7. Reproducibility and trust	29
7.1. Trust labels and correspondence	29
7.2. Replay and continuous integration	30
7.3. Relation to prior computations	30
7.4. Use of generative AI	31
8. Outlook	31
Appendix A. The componentwise saturation lemmas	31
Lemma B	31
Lemma C	31
Lemma D	32
Lemma E and Lemma E'	32
Proposition F (factorwise form)	32
Appendix B. Old and new components in cyclic p -towers	32
Appendix C. Characters of G_n and Frobenius orbits	33
Appendix D. Determinant normalisation and constants	33
Citation note	37
Appendix E. The scaling theorem	37
Appendix F. Artifact index	41
References	42

1. THE COMPUTATIONAL PROBLEM

This paper is organised around a computational problem. Fix a layer n of the cyclotomic $\mathbb{Z}_2$ -extension of $\mathbb{Q}$ and an odd prime ℓ , or a congruence class

of odd primes: decide $\ell \nmid k_n$, and produce a record of the decision that can be checked by machine, independently of the search that produced it. The mathematical core is a componentwise criterion (Theorem 1.3), an instance of an abstract saturation–height bound (Theorem 1.1); the computational deliverables are (i) enclosures of its constants C_n with exact outward endpoints, produced and certified by the algorithms of Section 4, (ii) for primes below the thresholds, a family of finite certificates whose acceptance by a read-only verifier implies the exclusion — soundness is a theorem (Section 3), the run is reported in Section 5 — and (iii) a trust architecture that assigns every claim a label stating what checked it, with kernel-checked discrete cores in Lean 4 (Section 7.1). Section 6 runs the same design in the cyclotomic $\mathbb{Z}_3$ -tower, where the covolume of the relevant lattice is computed exactly by a character product; Section 7 records how the package is replayed. Displayed decimals are never load-bearing: every proved numerical claim in the paper reduces to an integer comparison or to a containment between certified intervals.

1.1. Weber’s problem and its known controls. Let $B_n = \mathbb{Q}(X_n)$, $X_n = 2 \cos(2\pi/2^{n+2})$, be the layer of degree 2^n in the cyclotomic $\mathbb{Z}_2$ -extension of $\mathbb{Q}$, h_n its class number and $k_n = h_n/h_{n-1}$. Weber showed that h_n is odd; Weber’s class number problem asks whether $h_n = 1$ for all n . It is known that $h_n = 1$ for $n \leq 6$ (van der Linden for $n \leq 5$; Miller [10] Thm. 2.1 for $n = 6$), and Miller [10] Thm. 2.2 gives $h_7 = 1$ under the generalized Riemann hypothesis, using in its last step the unconditional exclusion of all $\ell < 10^9$ by Fukuda and Komatsu [2].

Three kinds of control on the odd primes dividing h_n are in the literature, and they fix different variables.

- (i) *Fix ℓ , climb n .* Washington [18, 19]: the exact power of ℓ dividing h_n is constant for large n (the case $p = 2$ is [18]); the argument is effective in the sense that a computable stabilization bound can be extracted from it ([19]); no numerical specialization of that bound is used here.
- (ii) *Fix a congruence class of ℓ , all n above a threshold.* Horie [4, 7] ($\ell \not\equiv \pm 1 \pmod{8}$), Horie–Horie [8] (explicit thresholds Θ per decomposition field, with the finite-layer clause “ $\ell \nmid h_{n_0}$ ”), Fukuda–Komatsu [2] ($\ell < 10^9$ or $\ell \not\equiv \pm 1 \pmod{32}$), Morisawa–Okazaki [13] ($\ell > G(2, s, f)$; $\ell \not\equiv \pm 1 \pmod{64}$).
- (iii) *Fix the layer.* Kashio–Yoshizaki [9] describe k_n through a saturation criterion on the irreducible factors of $x^{2^{n-1}} + 1$ over $\mathbb{F}_\ell$, prove an unconditional trace floor $33 \cdot 2^n$ on the relative units for $n \geq 3$, and exclude many primes at $n = 7$ conditionally on their Conjecture 2.2.

The present paper works at a fixed layer with all odd ℓ at once. Its criterion (Theorem 1.3) is componentwise: it applies Blichfeldt’s theorem [1] to the lattice of one saturation component and compares against the height floor of [13]. Where the criterion does not apply, the same components are searched

for finite witnesses; the result is a certificate family whose soundness is a theorem relative to the same literature inputs (Section 3).

The criterion is one instance of an abstract saturation–height bound (Theorem 1.1) with three inputs: a *carrier* (a saturation statement producing an ℓ -th root of a Galois-module element), a *floor* (a lower bound on the height of such roots) and a *lattice* (Blichfeldt’s theorem on the logarithmic image). The $\mathbb{Z}_2$ -tower supplies the carrier through Kashio–Yoshizaki, the floor through [13] and the lattice through Dirichlet’s unit theorem; the cyclotomic $\mathbb{Z}_3$ -tower supplies the same three inputs through Horie’s saturation lemma, the relative-norm-one height floor of [13] (the Horie unit has relative norm 1) and the exact covolume of the Horie-unit lattice, which is computed by a character product and is positive at every layer (Section 6). In the $\mathbb{Z}_3$ -tower this line of study goes back to Morisawa [11], who bounded the problem through the Mahler measure of the Horie unit. One theorem, two towers; the layer-fixed constants obtained in the second tower lie below the uniform-in- n bound of [13] in both congruence classes at the layer of degree 81.

1.2. Statement of results. Throughout, $n \geq 2$, $m = 2^{n-1}$, $q = 2^{n+2}$, $G_n = \text{Gal}(B_n/\mathbb{Q})$, σ the generator $X_n \mapsto 2 \cos(3 \cdot 2\pi/q)$, $\tau = \sigma^m$ the generator of $\text{Gal}(B_n/B_{n-1})$, E_n the units of B_n , $RE_n^+ = \{\varepsilon \in E_n : N_{n/n-1} \varepsilon = 1\}$, $\varepsilon_n = (X_n + 1)/(X_n - 1)$, and $A_n = \langle \pm 1, \varepsilon_n \rangle_{\mathbb{Z}[G_n]}$ (all as in [9] Def. 2.1). Heights are those of [13] Def. 2.2: $\text{ht}(\varepsilon) = (\sum_{i=1}^{2^n} (\log |\varepsilon_i|)^2)^{1/2}$ over all real embeddings. $L_n = \sqrt{2^n} \log(2 + \sqrt{5})$.

Theorem 1.1 (SH; componentwise saturation–height bound). *Let K be a totally real number field of degree N , $H : K^\times \rightarrow \mathbb{R}^N$ the logarithmic map over its real embeddings, $\text{ht} = |H(\cdot)|$, and $U \subseteq \mathcal{O}_K^\times$ any set of units. Let R be a free abelian group of rank m and $h : R \rightarrow \mathbb{R}^N$ an additive map with (Rank): h is injective; let D be the covolume of the lattice $h(R)$ in its $\mathbb{R}$ -span and $K_m = (2/\pi)^{m/2} \Gamma(2 + m/2)$. Let ℓ be an odd prime and $L \subseteq R$ a subgroup with (Index): $\ell R \subseteq L$ and $[R : L] = \ell^{m-d}$; let $L_0 > 0$. Assume*

- (Car) *for every $a \in L$ there is $u \in U$ with $\ell H(u) = h(a)$ (the scaled lattice vector $\ell^{-1}h(a)$ is carried by a unit of U);*
- (Floor) *every $u \in U$ with $H(u) \neq 0$ has $\text{ht}(u) \geq L_0$;*
- (Bl) *every lattice Λ of rank m in $\mathbb{R}^N$ contains $v \neq 0$ with $|v|^m \leq K_m \text{covol}(\Lambda)$ ([1] Thm. II, as [13] Thm. 2.7).*

Then $\ell^d \leq K_m D / L_0^m$.

Corollary 1.2 (SH-mod; the module form). *In the setting of Theorem 1.1 let moreover G be a finite abelian group acting on K by automorphisms, R a commutative quotient ring of $\mathbb{Z}[G]$ (free of rank m over $\mathbb{Z}$), U a G -stable subgroup of $\mathcal{O}_K^\times$, $A \subseteq U$ a G -stable subgroup containing ± 1 such that $A/\{\pm 1\}$ is a cyclic R -module generated by the class of $\varepsilon \in A$, $u_a = \varepsilon^a$ ($a \in R$), $h(a) = H(u_a)$, and $L = \pi^{-1}(M)$ the pullback of a nonzero ideal M of dimension d of the $\mathbb{F}_\ell$ -algebra $R/\ell R$. If*

(Sat) for every $a \in L \setminus \ell R$ there is $u \in U$ with $u^\ell = u_a$,
 then (Car) holds, and with (Floor) and (Bl) as in Theorem 1.1, $\ell^d \leq K_m D / L_0^m$.

Proof. $h(a) = H(u_a)$ is additive in a (WeberSH.H_pow: $H(u^k) = kH(u)$, and $u_{a+b} = \pm u_a u_b$ is invisible to H), so Theorem 1.1 applies once (Car) is verified for every $a \in L$.

Base branch ($a \in \ell R$). Write $a = \ell b$ with $b \in R$. Then $u_b \in A \subseteq U$ and $u_b^\ell = \pm u_a$, so $\ell H(u_b) = H(u_a) = h(a)$ (WeberSH.base_branch).

Saturation branch ($a \in L \setminus \ell R$). By (Sat) there is $u \in U$ with $u^\ell = u_a$; then $\ell H(u) = H(u_a) = h(a)$ (WeberSH.sat_root_log).

In both branches the carrier exists; Theorem 1.1 gives $\ell^d \leq K_m D / L_0^m$. (For a unit u of a totally real field, $H(u) = 0$ iff all conjugates are ± 1 iff $u = \pm 1$; so (Floor) in the form “ $u \neq \pm 1 \Rightarrow \text{ht}(u) \geq L_0$ ” is the same hypothesis.) $\square$

Theorem SH is Blichfeldt’s theorem applied to the scaled pullback lattice $\ell^{-1}h(L)$ of one simple component, against a height floor; its role is to separate the four inputs (Rank), (Index), (Car), (Floor) from the notation of the $\mathbb{Z}_2$ -tower. It is stated in exactly the form proved in Lean (WeberSH.theoremSH: h additive, a carrier for every $a \in L$, a floor on U); no group action and no module structure on the units enter the theorem. Corollary SH-mod is the form in which the $\mathbb{Z}_2$ -tower supplies the carrier: the base branch $a \in \ell R$ from the R -module structure of $A/\{\pm 1\}$ and the saturation branch from (Sat) (WeberSH.base_branch, WeberSH.sat_root_log). Theorem A is the specialisation of Corollary SH-mod to $K = B_n$, $R = R_n$, $U = RE_n^+$, $A = A_n$, $L = L_{f_0}$ (Section 2.6); Theorem 6.5 applies Theorem SH directly, with U the relative-norm-one units of the layer (Lemma 6.1).

Theorem 1.3 (A). *Let $n \geq 2$, ℓ an odd prime and $d_n(\ell) = \text{ord}_{2^n}(\ell)$. Let D_n be the covolume of the log-lattice $H_n(\mathbb{Z}^m)$ of A_n (Definition 2.2) and*

$$C_n = (2/\pi)^{m/2} \Gamma(2 + m/2) D_n / L_n^m.$$

If $\ell^{d_n(\ell)} > C_n$ then $\ell \nmid k_n$.

Theorem A is proved relative to the external literature results of Table 4, used as explicit hypotheses and never sharpened; no conjecture enters. Lean checks the discrete implication relative to the explicitly listed literature inputs (TheoremAInputs.theoremA_of_inputs, Section 7.1).

Proposition 1.4 (D). *With $S(\chi) = \sum_{g \in G_n} \chi(g) \log |g\varepsilon_n|$ and χ ranging over the m characters of G_n with $\chi(\tau) = -1$, equivalently the even Dirichlet characters of conductor exactly q :*

$$D_n = 2^{-m/2} \prod_{\chi} |S(\chi)| \quad \text{and} \quad D_n = 2^{m(n+1)/2+1} \prod_{\chi} |L(1, \chi)|.$$

The ingredients of Proposition D are classical: a group determinant and Washington's formula for $L(1, \chi)$. Its role here is to identify the exact normalization needed to turn the componentwise bound into an explicit and certifiable criterion, a product of L -values computable at every layer.

Corollary 1.5 (A'). $C_n = 2(4/\pi)^{m/2} \Gamma(2 + m/2) \prod_{\chi} |L(1, \chi)| / \log(2 + \sqrt{5})^m$.

Corollary 1.6 (T; order threshold). *For $B > 1$ put $f_n(B) = \max\{1, 1 + \lfloor \log C_n / \log B \rfloor\}$. If $\ell \geq B$ and $d_n(\ell) \geq f_n(B)$ then $\ell \nmid k_n$.*

Corollary 1.7 ($\widehat{A}$; computation-free criterion). *For even $q \geq 4$ put $U(q) = \frac{1}{4}(\log q + 2 \log 2) + \log(4q) / \sqrt{q}$, and*

$$\widehat{C}_n = 2(4/\pi)^{m/2} \Gamma(2 + m/2) (U(2^{n+2}) / \log(2 + \sqrt{5}))^m.$$

Then $C_n \leq \widehat{C}_n$ for every $n \geq 2$, and for every odd prime ℓ : $\ell^{d_n(\ell)} > \widehat{C}_n$ implies $\ell \nmid k_n$.

Corollary $\widehat{A}$ uses no computation of L -values: the only analytic input is the explicit bound $|L(1, \chi)| \leq U(q)$ of [15] Cor. 1 for even primitive characters of even conductor. It is coarse ($\widehat{C}_7 / C_7 \approx 4 \times 10^{22}$; Section 4) but applies at every layer without evaluating C_n .

Corollary 1.8 (explicit order threshold). *Put $k = 1 + m/2$ and*

$$T_n = \log 2 + \frac{m}{2} \log \frac{4}{\pi} + \frac{1}{2} \log(2\pi k) + k(\log k - 1) + \frac{1}{12k} + m \log \frac{U(2^{n+2})}{\log(2 + \sqrt{5})}.$$

Then $\log \widehat{C}_n \leq T_n$, and for every odd prime ℓ : $d_n(\ell) \log \ell > T_n$ implies $\ell \nmid k_n$.

Corollary 1.9 ($n = 7$).

$$C_7 = 1.7273421630363529579743237623519834054 \dots \times 10^{30}$$

(the 38 digits shown are a truncation of the certified enclosure of Section 4, truncation error $< 10^{-7}$; no other error is attached to the displayed digits). Hence $\ell \nmid h_7/h_6$ whenever $\ell \equiv 1 \pmod{128}$ and $\ell > 1727342163036353095979941756929$, or $\ell \equiv 63, 65, 127 \pmod{128}$ and $\ell > 1314283897427172$.

Theorem 1.10 (S0; abstract scale law). *Let $t \geq 2$ and let $J_t = \{a \in R_n : u_a \equiv 1 \pmod{2^t}\}$ where $u_a = \prod_i \sigma^i(\varepsilon_n)^{a_i}$, and put $L_{n,t} := \sqrt{2^n} \operatorname{arcsinh}(2^t)$. Assume (i) $J_t = 2^{t-1} R_n$. Then $\Lambda_{f,t} := \ell^{-1} H_n(L_f \cap J_t) = 2^{t-1} \Lambda_f$ for every component f , and the depth- t variant of Theorem A (Blichfeldt on $\Lambda_{f,t}$ against the depth- t floor $L_{n,t}$ of Lemma E.2, applied to the unit that carries the short vector) has a stricter sufficient condition than Theorem A, i.e. the resulting direct filtered-lattice criterion is strictly weaker, because $(2 + \sqrt{5})^{2^{t-1}} > 2^t + \sqrt{4^t + 1}$ for every $t \geq 2$.*

Corollary 1.11 (S1). *Hypothesis (i) holds for $t = 2, 3, 4$ at $n = 7$ (exact rank certificates). Hence the direct filtered-lattice routes at depths 4, 8 and 16 (Blichfeldt on $\Lambda_{f,t}$ against $L_{n,t}$) are strictly weaker than Theorem A. No statement is made about other 2-adic refinements.*

Note. Hypothesis (i) at $n = 7$ is a certificate input **[C]** (exact rank certificates).

Whether $J_t = 2^{t-1}R_n$ for $t \geq 5$ is not proved here; no claim is made about deeper depths. Write $\bar{T}_n := 17 \cdot 2^n$ for $n = 2$ and $\bar{T}_n := 33 \cdot 2^n$ for $n \geq 3$: the trace floors $\text{Tr}(\varepsilon^2) \geq \bar{T}_n$ on $RE_n^+ \setminus \{\pm 1\}$ of [14] Prop. 6.6 (every $n \geq 2$) and [9] Thm. 2.3 (stated for $n \geq 3$). The verifier recomputes $\bar{T}_n$ from n and accepts a T line only if $T < \bar{T}_n$ (Section 3).

Theorem 1.12 (Certificate soundness). *If the read-only verifier of Section 3 returns EXCLUDED for (n, ℓ) , then $\ell \nmid k_n$.*

Theorem 1.13 (Family). *Let $\mathcal{P}$ be the set of the first 1000 primes ℓ with $\ell > 10^9$ and $\ell \equiv 65 \pmod{128}$; $\min \mathcal{P} = 1000000321$, $\max \mathcal{P} = 1001287361$, and $\mathcal{P}$ is the list of [9] §4.2; equivalently, $\mathcal{P} = \{\ell \text{ prime} : 10^9 < \ell \leq 1001287361, \ell \equiv 65 \pmod{128}\}$. For every $\ell \in \mathcal{P}$ one has $\ell \nmid h_7/h_6$.*

The proof of Theorem 1.13 is one certificate per prime (32 components each) accepted by the verifier; Theorem 1.12 converts acceptance into exclusion. [9] §4.2 excludes the same 1000 primes conditionally on Conjecture 2.2; here no conjecture is used.

Remark 1.14 (numerical observation, not a theorem). For $2 \leq n \leq 9$ the computed value of $\sum_{\chi} \log |L(1, \chi)|$ lies in $[-0.37, 0.03]$ (Table 5); in this range the Gamma factor of Cor. A' dominates C_n numerically. Nothing is claimed about the L -product for $n \geq 10$.

1.3. Shape of the criteria. Tables 1 and 2 record, for each control, which variable is fixed and which moves, whether the statement is uniform in n , effective, componentwise, certificate-separated, and whether GRH enters. All rows were resolved from the primary texts (the verbatim quotations are in the repository file `theory/PHASE_MINUS1_INTEGRATED_AUDIT_R10.md`, part (c)). What each result of Sections 1.2 and 6 rests on, and which prior bound it is measured against, is tabulated in the repository ([NOVELTY], Appendix F).

The criterion of Theorem A is closest in shape to [9] (layer-fixed, componentwise) and closest in engine to [13] (Blichfeldt against the height floor), with two changes: the lattice is that of a saturation component $\ell^{-1}H_n L_f$ rather than Horie's ideal lattice, and its covolume is evaluated exactly (Prop. D) rather than bounded analytically.

1.4. The four surviving classes at $n = 7$. By [2] and [13] Cor. B, every odd prime ℓ with $\ell < 10^9$ or $\ell \not\equiv \pm 1 \pmod{64}$ satisfies $\ell \nmid h_n$ for all n . Above 10^9 the classes $\ell \equiv 1, 63, 65, 127 \pmod{128}$ remain; these are exactly the classes with $\text{ord}_{128}(\ell) \leq 2$. In the notation of [13], $f = \text{ord}_4(\ell)$ and

work	fixed	moving	uniformity in n	GRH
Washington [18, 19]	prime ℓ	layer n	e_n eventually constant	no
Horie [4, 7], Horie–Horie [8]	class of ℓ	all n	above explicit Θ ; finite-layer clause below	no
Fukuda–Komatsu [2]	none	all n	$\ell < 10^9$ or $\ell \neq \pm 1$ (32)	no
Morisawa–Okazaki [13]	class $D(2, s, f)$	all n	$\ell > G(2, s, f)$; $\ell \neq \pm 1$ (64)	no
Kashio–Yoshizaki [9]	layer n	prime ℓ	layer-fixed; floor $33 \cdot 2^n$ ($n \geq 3$); $n = 7$ exclusions under Conj. 2.2	no
Miller [10]	layer n	–	$h_6 = 1$; $h_7 = 1$ under GRH	$n = 7$
present work	layer n , explicit C_n	prime ℓ	formula at every n	no

TABLE 1. Controls on the odd primes dividing h_n , part 1: what is fixed, what moves, uniformity in the layer, use of GRH.

work	effective	componentwise	certificate
Washington [18, 19]	in principle	no	no
Horie [4, 7], Horie–Horie [8]	yes	no	no
Fukuda–Komatsu [2]	computational	no	no
Morisawa–Okazaki [13]	yes (analytic covolume bound)	no	no
Kashio–Yoshizaki [9]	yes	yes	printed tables
Miller [10]	–	–	–
present work	yes	yes	yes, read-only verifier

TABLE 2. Controls on the odd primes dividing h_n , part 2: effectivity, componentwise structure, certificates.

$2^s \parallel \ell^f - 1$; Theorem A of [13] excludes $\ell \in D(2, s, f)$ for all n whenever $\ell > G(2, s, f)$, with $G(2, s, f) = [2(\sqrt{\pi}/(\sqrt{2} \log(2 + \sqrt{5})))^c ((c+2)/2)!]^{1/f}$, $c = 2^{s-1}$. Its proof ([13] §3.3) first obtains, at a fixed layer n , the same bound with c replaced by $d = 2^{\min(n, s)-1}$, and then substitutes $d \rightarrow c$ to make the statement uniform in n . [13] prints only the values $G(2, 6, 1) < 7.8 \times 10^{12}$ and $G(2, 6, 2) < 2.8 \times 10^6$; every other value of G below was evaluated by us from the displayed formula (our evaluation reproduces the two printed bounds: 7.716×10^{12} and 2.778×10^6). The range not covered by the comparison theorems listed in Table 3 is: none in class 65 ([13] is $170\times$ sharper there); $(1.315 \times 10^{15}, 4.520 \times 10^{16})$ in class 63; $(1.315 \times 10^{15}, G(2, s, 2))$ in class 127; and $(1.727 \times 10^{30}, G(2, s, 1))$ in class 1. Three different mechanisms are visible in Table 3. In class 65 ($s < n$) the published theorem of [13] is sharper than Cor. 1.9 by a factor 170. In class 63 ($s = n$) the $d \rightarrow c$ substitution costs

class 128	mod	(f, s)	$d_7(\ell)$	Cor. 1.9 thresh- old	[13] value	layer-7	[13] $G(2, s, f)$	published
65		(1, 6)	2	1.315×10^{15}	7.716×10^{12}		7.716×10^{12}	
63		(2, 7)	2	1.315×10^{15}	4.520×10^{16}		4.520×10^{16}	
127		(2, $s \geq 8$)	2	1.315×10^{15}	4.520×10^{16}		$\geq 4.778 \times 10^{41}$	
1		(1, $s \geq 7$)	1	1.727×10^{30}	2.043×10^{33}		$\geq 2.043 \times 10^{33}$	

TABLE 3. The classes $\ell \equiv \pm 1 \pmod{64}$ at $n = 7$. “Layer-7 value” is the bound of [13] §3.3 before the $d \rightarrow c$ substitution, evaluated with $d = 2^{\min(7, s)-1}$.

nothing, and Cor. 1.9 is sharper than the published $G(2, 7, 2)$ by the constant alone (a factor 34, coming from the exact covolume and the component lattice). In classes 127 and 1 ($s > n$) Cor. 1.9 is sharper both by the constant ($34\times$ and $1183\times$ against the layer-7 values) and by not paying the $d \rightarrow c$ substitution, which for $s \geq 8$ raises G by many orders of magnitude. For general n the same three regimes occur, with s compared to n ; the position of the range covered by Theorem A and not by [13] is determined by C_n , whose computed values for $2 \leq n \leq 9$ increase with n (Table 5); nothing is claimed about C_n for $n \geq 10$.

The window $10^9 < \ell \leq 7.716 \times 10^{12}$ in class 65 is reached by none of the uniform theorems compared in Table 3; [9] §4.2 covers its first 1000 primes conditionally on Conjecture 2.2. Theorem 1.13 covers the same 1000 primes unconditionally. In the tested 1000-prime interval, ordinary certificate generation required 8–20 seconds per prime (one prime needed an escalation); no complexity or success-rate claim is made outside this interval. The window itself contains an estimated 4×10^9 primes (prime number theorem, equidistribution in the 64 classes mod 128); nothing here amounts to an exhaustive treatment of the window.

2. THE COMPONENTWISE CRITERION

2.1. Old and new components of the class group.

Lemma 2.1. *Let L/K be a cyclic extension of number fields of degree p^k and $\ell \neq p$ a prime. Let $\iota : \text{Cl}(K) \rightarrow \text{Cl}(L)$ be extension of ideals and $N : \text{Cl}(L) \rightarrow \text{Cl}(K)$ the norm. Then $\text{Cl}(L)_\ell = \iota(\text{Cl}(K)_\ell) \oplus \ker(N)_\ell$, ι is injective on $\text{Cl}(K)_\ell$, and $v_\ell(h_L) > v_\ell(h_K)$ if and only if $\ker(N)$ has an element of order ℓ . (In the Weber tower $k_n = h_n/h_{n-1}$ is an integer by the index formula of [9], and $v_\ell(h_n) > v_\ell(h_{n-1})$ is then $\ell \mid k_n$.)*

The proof is in Appendix B. Applied to B_n/B_{n-1} ($p = 2$) it shows that $k_n = h_n/h_{n-1}$, and not h_n , is the natural object at a fixed layer, and that the odd part of the “new” class group is $\ker(N)$. [9] Eq. (17), $k_n = [RE_n^+ : A_n]$, is the description of this quotient by relative units that Theorem A uses.

2.2. The exponent ring, the log map and the components.

Definition 2.2. $R_n = \mathbb{Z}[x]/(x^m + 1)$, with x acting on A_n through σ : for $a = \sum_{i < m} a_i x^i$ put $u_a = \prod_{i < m} \sigma^i(\varepsilon_n)^{a_i} \in A_n$. Let $\lambda_j = \log |\sigma^j \varepsilon_n|$ ($j \bmod 2m$); $\lambda_{j+m} = -\lambda_j$ because $N_{n/n-1} \varepsilon_n = 1$. The log map is $H_n : \mathbb{Z}^m \rightarrow \mathbb{R}^{2m}$, $(H_n a)_j = \sum_{i < m} a_i \lambda_{i+j}$, the log vector of u_a over the $2m = 2^n$ real embeddings. For an odd prime ℓ and an irreducible factor f of $x^m + 1$ over $\mathbb{F}_\ell$ put $w = (x^m + 1)/f$, $M_f = w \cdot \mathbb{F}_\ell[x]/(x^m + 1)$, and $L_f = \pi^{-1}(M_f) \subseteq R_n$ where $\pi : R_n \rightarrow R_n/\ell R_n = \mathbb{F}_\ell[x]/(x^m + 1)$. The lattice of the component f is $\Lambda_f = \ell^{-1} H_n(L_f)$.

By [9] §4, $A_n/\{\pm 1\}$ is free abelian on the classes of $\sigma^i(\varepsilon_n)$, $0 \leq i < m$, so $a \mapsto u_a$ identifies R_n with $A_n/\{\pm 1\}$ as $\mathbb{Z}[G_n]$ -modules. [9] define $A_n^{1/\ell} = \{x \in \mathbb{R} : x^\ell \in A_n\}$ (real ℓ -th roots) and identify

$$(1) \quad R_n/\ell R_n = \mathbb{F}_\ell[x]/(x^m + 1) \xrightarrow{\sim} A_n^{1/\ell}/A_n, \quad \bar{a} \mapsto r_a A_n, \quad r_a := \prod_{i < m} \sigma^i(\varepsilon_n)^{a_i/\ell}$$

where r_a is the real ℓ -th root of u_a ; this is well defined on classes mod ℓ because changing a by an element of ℓR_n multiplies r_a by an element of A_n . The map $M_f \rightarrow A_n^{1/\ell}/A_n$ is the restriction of (1) to the subspace M_f .

2.3. Saturation.

Lemma 2.3 (E; saturation chain). *Suppose $\ell \mid k_n$. Then there is an irreducible factor f_0 of $x^m + 1$ over $\mathbb{F}_\ell$ such that for every $a \in L_{f_0}$ the real formal root r_a lies in RE_n^+ , and its log vector is exactly $\ell^{-1} H_n a$.*

Proof. Step 1 (the index). $\ell \mid k_n = [RE_n^+ : A_n]$ ([9] Eq. (17), an exact equality).

Step 2 (types). Write $[r_a] := r_a A_n$ for the coset of the real formal root $r_a \in A_n^{1/\ell}$ (a number) in the group $A_n^{1/\ell}/A_n$; the identification (1) sends $\bar{a} \in R_n/\ell R_n$ to $[r_a]$, and it is well defined because replacing a by $a + \ell b$ multiplies r_a by $u_b \in A_n$ (the real ℓ -th root is unique since ℓ is odd: `WeberRoots`, `odd_pow_injective`, `WeberRoots.root_change_rep`, `WeberRoots.coset_eq_of_change_rep`). Since $A_n \subseteq RE_n^+$ ([9] Def. 2.1), the set $\{[\varepsilon] : \varepsilon \in RE_n^+ \cap A_n^{1/\ell}\}$ is a subgroup of $A_n^{1/\ell}/A_n$; it is what [9] write RE_n^+/A_n .

Step 3 (the published input). [9] Prop. 4.1 supplies an irreducible f_0 with $\{g \cdot \varepsilon_n : g \in \mathbb{Z}[x], g \bmod (x^m + 1) \in M_{f_0}\} \subseteq RE_n^+/A_n$, where $g \cdot \varepsilon_n$ is the real root r_g and the set is a set of cosets; in the identification (1) this says that every coset $[r_a]$ with $\bar{a} \in M_{f_0}$, i.e. with $a \in L_{f_0}$, belongs to the subgroup of Step 2: $[r_a] = [\varepsilon]$ for some $\varepsilon \in RE_n^+ \cap A_n^{1/\ell}$. (We do not say that a coset is a subset of RE_n^+/A_n ; membership of the coset in the image of $RE_n^+ \cap A_n^{1/\ell}$ is the statement.)

Step 4 (from the coset to the number). $[r_a] = [\varepsilon]$ means $r_a = \varepsilon c$ with $c \in A_n$. Since $A_n \subseteq RE_n^+$ and RE_n^+ is a group, $r_a \in RE_n^+$. This holds for every $a \in L_{f_0}$; if a is replaced by $a + \ell b$, r_a is multiplied by $u_b \in A_n \subseteq RE_n^+$, so membership in RE_n^+ is preserved, as it must be since $\ell R_n \subseteq L_{f_0}$.

Step 5 (the log vector). r_a is the real ℓ -th root of u_a , so $\ell \log |\sigma^j r_a| = \log |\sigma^j u_a| = (H_n a)_j$ for every real embedding σ^j , $j < 2m$: the log vector of r_a is $\ell^{-1} H_n a$. The base branch $a \in \ell R_n$ is Lemma E'. $\square$

Lemma 2.4 (E'; the base case $a \in \ell R_n$). *For $a = \ell b$, $b \in R_n$, the vector $\ell^{-1} H_n a = H_n b$ is the log vector of the unit $u_b \in A_n \subseteq RE_n^+$, and $u_b = \pm 1$ if and only if $b = 0$.*

Proof. The identity $\ell^{-1} H_n(\ell b) = H_n b$ is linear algebra; $u_b \in A_n \subseteq RE_n^+$ by [9] Def. 2.1; and $u_b = \pm 1$ forces $b = 0$ by the freeness of $A_n/\{\pm 1\}$ ([9] §4). $\square$

Lemma E' is the reason the certificate verifier of Section 3 demands $a \notin \ell R_n$ only as a bookkeeping condition: both branches lead to a relative unit, as in the proof of Theorem A below.

2.4. The height floor.

Lemma 2.5 (A; bridge). *If $u \in RE_n^+ \cap E_{n-1}$ then $u = \pm 1$.*

Proof. Recall from the setting (notation section of the paper; Blueprint definition “setting”) that $G_n = \langle \sigma \rangle$ is cyclic of order 2^n and $\tau = \sigma^{2^{n-1}}$ is its unique element of order 2; B_{n-1} is the fixed field of τ , so $\text{Gal}(B_n/B_{n-1}) = \{1, \tau\}$ and the relative norm is $N_{n/n-1}(x) = x \cdot \tau(x)$ for $x \in B_n$. By definition $RE_n^+ = \{\varepsilon \in E_n : N_{n/n-1}\varepsilon = 1\}$, and $E_{n-1} = E_n \cap B_{n-1}$ is the set of units fixed by τ .

Let $u \in RE_n^+ \cap E_{n-1}$. Because $u \in E_{n-1}$ we have $\tau(u) = u$; because $u \in RE_n^+$ we have $u \cdot \tau(u) = 1$. Substituting the first relation into the second gives $u \cdot u = u^2 = 1$. (This is `bridge_fixed_norm_sq`, stated for an arbitrary abelian group G with a homomorphism $\tau : G \rightarrow G$: from $\tau u = u$ and $u \tau u = 1$ one obtains $u^2 = 1$; it is applied with $G = E_n$.)

In the field B_n the identity $u^2 = 1$ reads $(u-1)(u+1) = 0$, and a field has no zero divisors, so $u = 1$ or $u = -1$ (`torsion_pm_one`; only the field axioms of B_n are used, not that B_n is totally real). Hence $RE_n^+ \cap E_{n-1} \subseteq \{\pm 1\}$. Conversely ± 1 are units of B_{n-1} with $N_{n/n-1}(\pm 1) = (\pm 1)^2 = 1$, so in fact $RE_n^+ \cap E_{n-1} = \{\pm 1\}$; only the inclusion is used later. $\square$

Lemma 2.6 (A^+ ; floor for relative units $\neq \pm 1$). $u \in RE_n^+ \setminus \{\pm 1\} \Rightarrow \text{ht}(u) \geq L_n$.

Proof. Let $u \in RE_n^+ \setminus \{\pm 1\}$. By Lemma 2.5, $u \notin E_{n-1}$, and $N_{n/n-1} u = 1$ by the definition of RE_n^+ . These are exactly the two hypotheses of [13] Lemma 2.5(1) (§3.3 of [13], p. 817: $\varepsilon \in E_n \setminus E_{n-1}$ with $\text{Nr}_{B_n/B_{n-1}}(\varepsilon) = 1$), whose conclusion is $\text{ht}(u) \geq \sqrt{2^n} \log(2 + \sqrt{5}) = L_n$. $\square$

Both hypotheses of [13] Lemma 2.5(1) — $u \notin E_{n-1}$ and relative norm exactly 1 — are load-bearing, and Lemma A is what supplies them.

2.5. The lattice of a component.

Lemma 2.7 (B; equal degrees). *All irreducible factors of $x^m + 1$ over $\mathbb{F}_\ell$ have degree $d_n(\ell) = \text{ord}_{2^n}(\ell)$. At $n = 7$: $d = 1$ for $\ell \equiv 1 \pmod{128}$ and $d = 2$ for $\ell \equiv 63, 65, 127 \pmod{128}$.*

Lemma 2.8 (C; index). $[R_n : L_f] = \ell^{m-d_f}$, where $d_f = \deg f$.

Lemma 2.9 (D; rank and covolume). H_n is injective, $H_n(\mathbb{Z}^m)$ is a discrete subgroup of $\mathbb{R}^{2m}$, hence a lattice of rank m with covolume $D_n = 2^{m/2} |\det W_n| > 0$, where $(W_n)_{j,i} = \lambda_{i+j}$ ($0 \leq i, j < m$). Consequently Λ_f is a lattice of rank m and covolume D_n/ℓ^{d_f} .

Proofs: Appendix A. Lemma D uses two inputs and needs both: injectivity comes from the freeness of $A_n/\{\pm 1\}$, discreteness from Dirichlet's unit theorem; injectivity alone does not give real rank m .

2.6. Proof of Theorem A.

Proof of Theorem 1.1. Covolume. Put $\Lambda := \ell^{-1}h(L) = \{\ell^{-1}h(a) : a \in L\}$. By (Rank) h is an injective $\mathbb{Z}$ -linear map $R \rightarrow \mathbb{R}^N$, so $h(L)$ is a sublattice of $h(R)$ of index $[R : L] = \ell^{m-d}$ (Index) and has covolume $\ell^{m-d}D$ in the common m -dimensional $\mathbb{R}$ -span; the map $v \mapsto \ell^{-1}v$ on that span multiplies m -dimensional volumes by ℓ^{-m} , so $\text{covol } \Lambda = \ell^{m-d}D/\ell^m = D/\ell^d$.

Short vector. By (Bl) there is $a \in L$ with $v := \ell^{-1}h(a) \neq 0$ and $|v|^m \leq K_m \text{covol } \Lambda = K_m D/\ell^d$.

Carrier and floor. By (Car) there is $u \in U$ with $\ell H(u) = h(a)$, i.e. $H(u) = v \neq 0$, and (Floor) gives $|v| = \text{ht}(u) \geq L_0$ (WeberSH.short_vector_ge_floor).

Hence $L_0^m \leq |v|^m \leq K_m D/\ell^d$, i.e. $\ell^d \leq K_m D/L_0^m$ (WeberSH.theoremSH; contrapositive WeberSH.theoremSH_contra). The covolume computation is the natural-language part of this proof; the short-vector, carrier and floor steps are the Lean theorem verbatim, with $h(a) = H(\iota(a))$ for the map $\iota : \mathbb{Z}^m \rightarrow U$ of the Lean statement. Nothing about R beyond additivity of h , nothing about U beyond (Car) and (Floor), and no group action is used: these enter only through how L , d and (Car) are produced in applications (Corollary 1.2 for the $\mathbb{Z}_2$ -tower; Lemma 6.1 for the $\mathbb{Z}_3$ -tower). $\square$

Proposition 2.10 (F; factorwise bound). *If $\ell \mid k_n$ and f_0 is the saturation component of Lemma 2.3, then $\ell^{d_{f_0}} \leq C_n$.*

Proof. Apply Corollary 1.2 (the module form of Theorem 1.1) with $K = B_n$, $N = 2^n$, $G = G_n$, $R = R_n = \mathbb{Z}[x]/(x^m + 1)$ (the quotient of $\mathbb{Z}[G_n]$ by $1 + \tau$, which acts on A_n as the norm to B_{n-1} , trivial on $RE_n^+ \supseteq A_n$), $U = RE_n^+$, $A = A_n$, $\varepsilon = \varepsilon_n$ (so u_a is the unit $\prod_i \sigma^i(\varepsilon_n)^{a_i}$ of the setting; $A_n/\{\pm 1\}$ is free on $\sigma^i \varepsilon_n$, [9] §4, hence cyclic over R_n), (Rank) by Lemma 2.9, $D = D_n$, $M = M_{f_0}$ of dimension $d_{f_0} = \deg f_0$, $L = L_{f_0}$ with (Index) by Lemma 2.8, and $L_0 = L_n$. Hypothesis (Sat) is Lemma 2.3: for $a \in L_{f_0} \setminus \ell R_n$ the real root $r_a \in RE_n^+$ satisfies $r_a^\ell = u_a$ (the base branch $a \in \ell R_n$ of Corollary 1.2

is Lemma 2.4). Hypothesis (Floor) is Lemma 2.6 (Lemma 2.5 with [13] Lemma 2.5(1)). Hypothesis (Bl) is [1] Thm. II in the form [13] Thm. 2.7. Corollary 1.2 gives $\ell^{d_{f_0}} \leq K_m D_n / L_n^m = C_n$.

Unfolded, for the reader who wants the argument in place: $\Lambda_{f_0} = \ell^{-1}H_n(L_{f_0})$ has rank m and covolume D_n/ℓ^{d_0} (Lemma 2.9); if $\ell^{d_0} > C_n$, Blichfeldt gives $0 \neq v = \ell^{-1}H_n a \in \Lambda_{f_0}$ with $|v|^m \leq (2/\pi)^{m/2} \Gamma(2 + m/2) D_n / \ell^{d_0} \leq C_n L_n^m / \ell^{d_0} < L_n^m$; if $a \in \ell R_n$, Lemma 2.4 makes v the log vector of $u_b \in RE_n^+ \setminus \{\pm 1\}$, if $a \notin \ell R_n$, Lemma 2.3 makes v the log vector of $r_a \in RE_n^+ \setminus \{\pm 1\}$; Lemma 2.6 gives $\text{ht} \geq L_n$, contradicting $|v| < L_n$. $\square$

Proof of Theorem 1.3. Suppose $\ell \mid k_n$. Proposition 2.10 gives $\ell^{d_{f_0}} \leq C_n$ for the saturation component f_0 , and Lemma 2.7 gives $d_{f_0} = d_n(\ell)$; hence $\ell^{d_n(\ell)} \leq C_n$. Contrapositively, $\ell^{d_n(\ell)} > C_n$ implies $\ell \nmid k_n$. The discrete core of this argument (saturation exists, every lattice vector obeys the Blichfeldt bound in the power- $m/2$ form, saturation forces height $\geq L$, hence $C < \ell^d$ is impossible) is the Lean theorem `TheoremAInputs.theoremA_of_inputs`. $\square$

The discrete core of this argument — “if some component is saturated, if every vector of its lattice satisfies the Blichfeldt bound in the power- $m/2$ form, and if saturation forces height $\geq L$, then $C < \ell^d$ is impossible” — is the Lean theorem `TheoremAInputs.theoremA_of_inputs`, with the three inputs as fields of a structure (Section 7.1). Cor. T is `theoremA_threshold`; the integer comparison of Cor. 1.9 is `c7_deg2_instance` and `deg1/deg2_threshold_valid`.

3. CERTIFICATE SOUNDNESS

The central computational object of the paper is a pair: an untrusted generator that searches for witnesses, and a read-only verifier whose acceptance is what the theorems consume. The division is asymmetric by design. The generator may use any heuristics and any software, and nothing it produces is trusted; the verifier performs no search, recomputes everything it checks from the definitions, and is small enough for its specification to be stated in full below. Soundness — verifier acceptance implies the divisibility exclusion — is Theorem 1.12, proved relative to the explicit literature inputs of Table 4; the specification itself enters the Lean statement as an explicit hypothesis (`exclusion_of_checker_spec`).

3.1. Pipeline: untrusted generator, read-only verifier. A generator (untrusted; Appendix F) searches each component lattice Λ_f for a vector $a \in L_f \setminus \ell R_n$ with either $\text{ht}(\ell^{-1}H_n a) < L_n$ (RHO route) or $T(a) = \sum_{j < 2m} \exp(2y_j) < \bar{T}_n$ with $y = \ell^{-1}H_n a$ (T route; $\bar{T}_n = 33 \cdot 2^n$ for $n \geq 3$ by [9] Thm. 2.3 and $17 \cdot 2^n$ for $n = 2$ by [14] Prop. 6.6; the verifier recomputes $\bar{T}_n$ from n and rejects a header that claims any other threshold),

input	statement used	where it enters
[9] Eq. (17)	$k_n = [RE_n^+ : A_n]$, exact	Lemma E, step 1
[9] §4	$A_n/\{\pm 1\}$ free on $\sigma^i \varepsilon_n$; identification (1)	Def. 2.2, Lemmas D, E'
[9] Prop. 4.1 with display (23)	$\ell \mid k_n \Rightarrow$ some f_0 with $[r_a] = [\varepsilon]$, $\varepsilon \in RE_n^+ \cap A_n^{1/\ell}$, for every $a \in L_{f_0}$	Lemma E
[9] Def. 2.1	$A_n \subseteq RE_n^+$	Lemmas E, E'
[13] Lemma 2.5(1)	$\text{ht}(u) \geq \sqrt{2^n} \log(2 + \sqrt{5})$ for $u \in E_n \setminus E_{n-1}$, $N_{n/n-1} u = 1$	proof of Thm. A
[13] Thm. 2.7 = [1] Thm. II	$ v ^2 \leq \frac{2}{\pi} \Gamma(2 + \frac{d}{2})^{2/d} \text{covol}^{2/d}$	proof of Thm. A
Dirichlet's unit theorem	log-image of E_n is discrete	Lemma D
[20] Chapter 4 ($L(1, \chi)$ formula; Gauss-sum modulus)	$L(1, \chi)$ formula, $ \tau(\chi) = \sqrt{q}$	Prop. D(ii) only
[15] Cor. 1 ($h = 1, k = 2$)	$ L(1, \chi) \leq \frac{1}{4}(\log q + 2 \log 2) + \frac{\log 4q}{\sqrt{q}}$, χ even primitive, $2 \mid q \geq 4$	Cor. $\hat{A}$ only
[16]	$k! \leq \sqrt{2\pi k} (k/e)^k e^{1/(12k)}$	Cor. 1.8 only
[12] Lemma 1.3 ([6], [3])	$\ell \mid h_{3,n}/h_{3,n-1} \Rightarrow \exists \mathcal{L} \mid \ell$ of $\mathbb{Q}(\zeta_{3r})$: $\eta_n^{\alpha\sigma} \in E_{3,n}^\ell$ for all $\alpha \in \ell \mathcal{L}^{-1}$	Thm. 6.5 only
[12] Thm. 2.2 ([17])	$M(\varepsilon) \geq ((1 + \sqrt{5})/2)^{\deg \varepsilon/2}$ for a totally real unit $\varepsilon \neq \pm 1$	Thm. 6.5 only

TABLE 4. Trust table: every literature input of Theorem A and Prop. D. Not used: [9] Conj. 2.2, Horie's lemma, any lattice search.

using LLL/BKZ and, when needed, exact SVP. Nothing produced by the generator is trusted.

Algorithm 3.1 (VERIFYCOMPONENTCERTIFICATE). A separate verifier (the family verifier of Appendix F, 80 lines, no search, no dependence on the generator) reads a witness file and recomputes from the definitions, in ball arithmetic at 256 bits, checking:

- (V1) the factorisation of $x^m + 1$ over $\mathbb{F}_\ell$, rejecting the file unless it is square-free with all factors of degree $\text{ord}_{2^n}(\ell)$ and unless the header fields $n, \ell, m, \deg, \# \text{components}, \bar{T}_n, L_n$ agree with the recomputation;
- (V2) the identity of each named factor by its full coefficient list;
- (V3) the exact membership $f \cdot a \equiv 0 \pmod{(\ell, x^m + 1)}$, and $a \notin \ell R_n$;
- (V4) the certified nonvanishing of some log coordinate;
- (V5) the route inequality with the ball's upper endpoint;
- (V6) agreement of the claimed value with the recomputed one to 10^{-6} .

It returns EXCLUDED only if every component carries at least one accepted line and no line fails. [C]

3.2. Soundness.

Proof of Theorem 1.12. Suppose $\ell \mid k_n$. [9] Prop. 4.1 gives a saturated component f_0 (Lemma E). The accepted line on f_0 names a vector $a \in L_{f_0} \setminus \ell R_n$ with $v = \ell^{-1} H_n a \neq 0$; by Lemma E, v is the log vector of $r_a \in RE_n^+$, $r_a \neq \pm 1$. A RHO line asserts $|v| < L_n$, contradicting Lemma A with [13] Lemma 2.5(1); a T line asserts $\sum_j \sigma^j(r_a)^2 = \sum_j e^{2y_j} < \bar{T}_n$, where $\bar{T}_n = 33 \cdot 2^n$ for $n \geq 3$ and $\bar{T}_n = 17 \cdot 2^n$ for $n = 2$ (the verifier recomputes $\bar{T}_n$ from n), contradicting the trace floor $\text{Tr}(r_a^2) \geq \bar{T}_n$ of [9] Thm. 2.3 for $n \geq 3$ and of [14] Prop. 6.6 for $n = 2$ (the uniform constant $33 \cdot 2^n$ would not give a contradiction at $n = 2$ for $68 \leq T < 132$). Hence $\ell \nmid k_n$. $\square$

The propositional core — saturation exists, each witness kind refutes saturation, every component carries a witness, hence no divisibility — is `WeberCert.exclusion_chain_direct`; the two refutation steps are `rho_witness_refutes_saturation` and `t_witness_refutes_saturation`, whose hypotheses are the two floors; and `exclusion_of_checker_spec` states the specification “verifier accepts $\Rightarrow$ every component has a witness of one of the two kinds” as an explicit hypothesis. The threshold $\bar{T}_n$ itself is the definition `WeberCert.barT` (Appendix F: $\bar{T}_2 = 68$, $\bar{T}_7 = 4224$, $\bar{T}_n \leq 33 \cdot 2^n$ for every n , strict at $n = 2$ where $\bar{T}_n = 17 \cdot 2^n$, and `t_witness_refutes_saturation_barT`, the T-refutation instantiated at $\bar{T}_n$). Lean does not verify the implementation of the verifier; it verifies the argument from what the verifier is specified to check. The trust label is therefore C+L with an F-core, not F.

4. EXACT EVALUATION OF THE CONSTANTS

4.1. Proof guide for Proposition D. The characters of G_n with $\chi(\tau) = -1$ are the even characters of $(\mathbb{Z}/q)^\times$ of conductor exactly q , determined by $\chi(3)$, which runs over all primitive 2^n -th roots of unity (Appendix C). Part (i) is the group determinant: the Gram matrix $H_n^T H_n$ is negacirculant, its eigenvectors are $(\omega^i)_{i < m}$ for $\omega^m = -1$, with eigenvalues $|S(\chi)|^2/2$ where $\omega = \chi(3)$. Part (ii) writes $\varepsilon_n = (1 - \zeta^3)^2(1 - \zeta^2)/((1 - \zeta)^2(1 - \zeta^6))$ with $\zeta = e^{2\pi i/q}$, drops the terms with even exponent by orthogonality against a character of conductor q , and applies Washington’s formula $|\sum_{a \in (\mathbb{Z}/q)^\times} \chi(a) \log |1 - \zeta^a|| = \sqrt{q} |L(1, \chi)|$ ([20] Chapter 4, the $L(1, \chi)$ formula together with $|\tau(\chi)| = \sqrt{q}$), obtaining $|S(\chi)| = |1 - \chi(3)| \sqrt{q} |L(1, \chi)|$ and $\prod_\chi |1 - \chi(3)| = \Phi_{2^n}(1) = 2$. Full proofs: Appendix D. $D_n > 0$ follows also from $L(1, \chi) \neq 0$ (Dirichlet), independently of Lemma D.

4.2. Numerical values. Table 5 gives C_n for $2 \leq n \leq 9$. Each $L(1, \chi)$ is enclosed in a rigorous ball through the digamma formula $L(1, \chi) = -q^{-1} \sum_{a=1}^q \chi(a) \psi(a/q)$, valid for every nontrivial $\chi \bmod q$ (from

$L(s, \chi) = q^{-s} \sum_a \chi(a) \zeta(s, a/q)$ at $s = 1$), evaluated in ball arithmetic at 500 bits; the products, the Gamma factor and the powers are ball operations, so each row is a rigorous enclosure ($[\mathbf{C}]$). At $n = 7$ three further routes are compared with the enclosure *as intervals* (cross-CAS audit script and result, Appendix F): the interval determinant of W_7 at 256, 512 and 1024 bits (rigorous; the four enclosures are nested consistently, the 1024-bit determinant interval, of width 10^{-139} , lying inside the digamma interval and the digamma interval inside the two coarser ones; the 256-bit upper endpoint is the origin of the frozen integer T_1), Washington’s Gauss-sum formula in PARI (200 bits; $\log D_7$ lies inside the 256-bit determinant interval; numerical audit, not part of the certificate), and a Magma evaluation with L -values at 40 digits (numerical audit: the printed Magma value agrees with the certified enclosure well beyond its working-precision-based conservative estimate $v(1 \pm 10^{-37})$, which intersects the enclosure; the length of that agreement is an observation and carries no accuracy claim—only the certified Sage/Arb enclosure carries a trust label). Agreement of leading digits and interval inclusion are reported separately; no tolerance is used. The digamma route is independent of Washington’s formula and of the determinant, so the agreement is a check of Prop. D(ii) itself.

n	m	$\log_{10} C_n$	$\sum_{\chi} \log L(1, \chi) $	enclosure
2	2	0.2285	-0.3673	ball, radius $\leq 2.8 \times 10^{-149}$
3	4	0.5475	-0.2388	ball, radius $\leq 3.1 \times 10^{-148}$
4	8	1.4268	-0.2242	ball, radius $\leq 1.8 \times 10^{-146}$
5	16	4.1589	+0.0233	ball, radius $\leq 4.8 \times 10^{-143}$
6	32	11.2900	-0.3177	ball, radius $\leq 4.4 \times 10^{-135}$
7	64	30.2374	-0.3550	ball, radius $\leq 2.2 \times 10^{-115}$; three further routes agree
8	128	77.5244	+0.0077	ball, radius $\leq 2.8 \times 10^{-67}$
9	256	190.5832	-0.0506	ball, radius $\leq 2.0 \times 10^{47}$

TABLE 5. The constant of Theorem A (Cor. A’): digamma-route enclosures, every row generated from the interval certificate (Appendix F); the radius column is an upper bound on the ball radius, the two numeric columns are midpoints rounded to four decimals.

Remark 4.1. The 2^n -powers cancel between D_n and L_n^m (Cor. A’). Any improvement of the geometry-of-numbers constant in Blichfeldt’s theorem would lower C_n multiplicatively at every layer at once. The behaviour of $\prod_{\chi} |L(1, \chi)|$ for large n is not studied here (Remark 1.14).

4.3. Algorithms and cost. The enclosures of Table 5 and of Table 8 are produced by the two procedures below. Both run entirely in ball arithmetic (Arb, through Sage’s `RealBallField` and `ComplexBallField` at a fixed working precision of p bits): every operation returns a midpoint–radius

ball containing the exact value, so each final enclosure is a rigorous bound, with the trust boundary of label [C] (Section 7.1). All stored bounds are exact endpoints rounded outward; decimal displays are derived from them and are never themselves load-bearing.

Evaluation of C_n (digamma route).

Algorithm 4.2 (CERTIFYCN). Input n ; put $q = 2^{n+2}$, $N = 2^n$, $m = 2^{n-1}$, $p = 500$.

- (1) *Characters.* Build the index table $j \mapsto \pm 3^j \bmod q$ once ($O(N)$ steps). The m even characters of conductor exactly q are $\chi_r(a) = \omega_r^{j(a)}$ with $\omega_r = e^{2\pi i(2r+1)/N}$, $0 \leq r < m$, the m primitive N -th roots of unity (Appendix C); the powers ω_r^j are precomputed as complex balls.
- (2) *L-values.* Compute $\psi(a/q)$ once for each odd $a < q$ (2^{n+1} digamma evaluations, shared by all characters), then $L(1, \chi_r) = -q^{-1} \sum_a \chi_r(a) \psi(a/q)$, a sum of 2^{n+1} ball products per character.
- (3) *Assembly.* $C_n = 2 (4/\pi)^{m/2} \Gamma(2 + m/2) \prod_r |L(1, \chi_r)| / \log(2 + \sqrt{5})^m$ as a single real ball.
- (4) *Endpoints.* Extract the exact rational endpoints of the ball; round them outward (lower endpoint down, upper endpoint up) to 160 significant decimal digits. The run asserts that the two endpoint strings differ, that the lower is strictly below the upper, and that their common leading-digit prefix has at least 38 digits; the displayed value is a truncated prefix of that certified prefix, so its only error is the stated truncation bound, and the ball radius is recorded as a separate field, never attached to the display.
- (5) *Frozen-threshold gate.* At $n = 7$ the run fails unless the upper endpoint is below both frozen integers of Cor. 1.9 (the integers of the Lean declarations `deg1_threshold_valid`, `deg2_threshold_valid`; they are compared against, never regenerated).

The ball-operation count of step (2) is $\Theta(4^n)$ ($m \cdot 2^{n+1} = 2^{2n}$ products), in addition to the 2^{n+1} digamma evaluations; the precomputed power tables of step (1) hold $m \cdot N = \Theta(4^n)$ complex balls, which is the peak memory of the implementation. No bit-complexity bound is claimed for the transcendental ball operations; measured costs are reported in Table 7. At $p = 500$ the certified decimal prefix has at least 143 digits at every layer $n \leq 9$ (the per-row radius bounds are fields of the constants certificate, Appendix F), far beyond the 38-digit display gate; the margin is deliberate, so that the certificate, not the working precision, is the binding constraint.

Exact covolume at $p = 3$ (two routes, one certificate).

Algorithm 4.3 (CERTIFYP3COVOLUME). Input $n \leq 5$; put $N = 3^n$, $q = 3^{n+1}$, $p = 4000$. For each $r \leq n$ write $c = 2 \cdot 3^{r-1}$.

- (1) *Log profile.* $\lambda_t = \log |\sigma^t \eta_n|$ for $t \in \mathbb{Z}/N$, each a real ball from the sine quotient defining η_n . Two structural gates: $\sum_t \lambda_t$ contains 0 (unit), and $\lambda_t + \lambda_{t+N/3} + \lambda_{t+2N/3}$ contains 0 for every t (relative norm 1, Lemma 6.1); either failure aborts the run.
- (2) *Transform.* $\hat{\lambda}_k = \sum_t \lambda_t e^{-2\pi i t k / N}$ by the naive N^2 sum in complex balls (at $N \leq 243$ no fast transform is needed, and each coefficient's radius then composes along a single sum of products). Gates: $\hat{\lambda}_k$ contains 0 for every $3 \mid k$ (the coset vanishing of Theorem 6.3, asserted, not assumed), and $\min_{3 \nmid k} |\hat{\lambda}_k|$ has a positive lower endpoint — the certified spectral-rank fact.
- (3) *Route one (Gram).* $M \in \mathbb{R}^{c \times N}$ with rows $H(\sigma^{3^{n-r}i} \eta_n)$, $i < c$; $D_{\text{gram}} = \sqrt{\det(MM^T)}$, the lower endpoint of the determinant asserted positive. Cost $O(c^2 N)$ products and an $O(c^3)$ elimination.
- (4) *Route two (character product).* $W_{n,r}(a) = N^{-1} \sum_{k \equiv a \pmod{3^r}} |\hat{\lambda}_k|^2$ and $D_{\text{dft}}^2 = 3^{3^r-1(2r-1)} \prod_{a \in (\mathbb{Z}/3^r)^\times} W_{n,r}(a)$ — the formula of Theorem 6.3 — in $O(N)$ operations once the transform is computed.
- (5) *Cross-gate.* $D_{\text{gram}}/D_{\text{dft}} - 1$ must contain 0; the enclosure of D is the intersection of the two routes' balls (both contain the true value, so the intersection does).
- (6) *Constants.* $C_{n,r}^{(3)} = (2/\pi)^{c/2} \Gamma(2 + c/2) D/L_{\text{rel}}^c$ and the comparator $G(3, s, f)$ of [13]; every improvement claim in Table 8 is an endpoint comparison (upper endpoint of one interval below lower endpoint of the other), never a comparison of midpoints.
- (7) *Certification.* The certificate [P3-CERT] is generated by [P3-GEN] as the outward hull of the producer balls and of an independent read-only recomputation, widened to twice its half-width, and stored as exact dyadic endpoints ($\text{sign} \cdot \text{mantissa} \cdot 2^{\text{exp}}$). The replay [P3-REPLAY] rebuilds every quantity from the definitions and requires containment of its recomputation in all 120 certified intervals; the twelve planted mutant certificates are rejected (Appendix F).

At fixed precision p , the naive transform uses $\Theta(N^2) = \Theta(9^n)$ complex-ball products ($N^2 = 59049$ at $n = 5$). For fixed r the Gram route uses $O(c^2 N + c^3)$ ball operations; at the worst case $r = n$, where $c = 2N/3$, this is $O(N^3) = O(27^n)$, so the Gram route, not the transform, determines the asymptotic ball-operation count of the two-route certification. No bit-complexity theorem is claimed for the transcendental ball operations; measured wall times in the pinned environments are reported in Table 7. At $p = 4000$ every row of Table 8 is an enclosure of radius below 10^{-800} .

Design: exact outward endpoints. Three consequences of storing exact dyadic outward endpoints rather than displayed decimals. First, the proved value and the printed value are separated by construction: a display string can be regenerated, truncated or mistranscribed without touching the certificate. Second, the verifier needs no state from the producer:

containment of a from-the-definitions recomputation in a stored interval is a complete check, so the producer is untrusted (Section 3). Third, the doubled hull provides fixed slack against the small last-bit differences observed between the pinned producer and replay environments; a recomputation not contained in a frozen interval is rejected rather than silently accepted, and no platform-independence theorem is claimed.

5. THE CERTIFIED COMPUTATION FOR THE THOUSAND PRIMES

5.1. The family. Theorem 1.13: $\mathcal{P}$ is defined as the first 1000 primes $\ell > 10^9$ with $\ell \equiv 65 \pmod{128}$; the list was regenerated independently and coincides element by element with the list produced by the generator, and its endpoints coincide with those printed in [9] §4.2. Table 6 summarises the certificates. Every component of every prime carries a T certificate; 31987 of the 32000 components also carry a RHO certificate, 13 are T-only. One prime (1000992193) required an escalated search (BKZ-60 and exact SVP) for one component.

Lemma 5.1 (KY1000 certificates). *For every $\ell \in \mathcal{P}$ the witness file for ℓ (one file per prime, `certificates/family/KY1000/`, SHA-256 in the manifest; Appendix F) is accepted by the frozen family verifier (SHA-256 `0f38bb0d...`) with verdict EXCLUDED: the verification ledger reads EXCLUDED 1000/1000; nine planted corruptions of a passing file are all REJECTED and the untouched $n = 2$ file is accepted (negative-control log and ledger, Appendix F).*

Proof of Theorem 1.13. The finite set $\mathcal{P}$. Define $\mathcal{P}$ as the set of the first 1000 primes $\ell > 10^9$ with $\ell \equiv 65 \pmod{128}$, i.e. the 1000 smallest such primes. This is a finite, explicitly enumerable set: the primes $\equiv 65 \pmod{128}$ above 10^9 form an increasing sequence, and $\mathcal{P}$ is its initial segment of length 1000. Its largest element is 1001287361, so $\mathcal{P} = \{\ell \text{ prime} : 10^9 < \ell \leq 1001287361, \ell \equiv 65 \pmod{128}\}$; the two descriptions agree because the second set has exactly 1000 elements and contains the first. The endpoints coincide with those printed in [9] §4.2.

Target-list certificate (claim KY1000_TARGET, label C). The shipped list (`sage/family_ky1000/KY1000_primes.txt`) is checked, by a script with no dependence on the generator, to have 1000 entries, distinct and ascending, every entry prime (deterministic primality test for this size), every entry $\equiv 65 \pmod{128}$ and $> 10^9$; the script then regenerates the first 1000 qualifying primes from 10^9 by a sieve and compares elementwise, and finally compares the list with the set of primes in the witness-file names. The target-list checker also rejects five planted corruptions of the list (one entry altered to a composite in the wrong class; one qualifying prime omitted and the list padded with the 1001st; the last entry duplicated; two neighbours swapped; one prime replaced by the next qualifying prime beyond 1001287361, which only the interval form catches; verifier steps 03b/03c, a control set separate

from the nine witness-file controls). This discharges the identification of the shipped list with the set $\mathcal{P}$ defined above.

Witness certificates (Lemma 5.1, label C). For every $\ell \in \mathcal{P}$ there is one witness file (SHA-256 in the manifest). The read-only verifier `scripts/family_verify.sage` (specification in the paper’s certificate section; frozen; SHA-256 0f38bb0d...) accepts each of them with verdict EXCLUDED: the verification ledger records EXCLUDED 1000/1000, every one of the $32 = 2^{7-2}$ components of every prime carrying an accepted T certificate (32000 T lines) and 31987 of them also an accepted RHO line; the largest accepted T value is $4164.4897 < \bar{T}_7 = 4224$. Nine planted corruptions of a passing file (wrong factor, wrong vector, vector in ℓR_n , altered threshold, altered header fields, ...) are REJECTED and the untouched $n = 2$ file is accepted (`sage/negctl_r13/negctl_r13.log`; ledger `certificates/negctl/negctl_ledger_r14.json`).

From EXCLUDED to $\ell \nmid h_7/h_6$ (Theorem 1.12, C+L, F-core). By Theorem 1.12, EXCLUDED for $(7, \ell)$ implies $\ell \nmid k_7 = h_7/h_6$: each accepted line is a nonzero vector of the component lattice refuting saturation of that component (RHO: height below the floor L_7 ; T: trace of the square below $\bar{T}_7$, [9] Thm. 2.3), and saturation of some component is forced by $\ell \mid k_7$ ([9] Prop. 4.1). Both refutations are instances of the same discrete core as Theorem A (`WeberCert.exclusion_chain_direct`); the literature inputs are those of the trust table of the paper (KY Eq. (17), KY Prop. 4.1, KY Def. 2.1, MO16 Lemma 2.5(1), Blichfeldt) plus the trace floor KY Thm. 2.3 / MO3 Prop. 6.6.

Scope. Nothing is claimed for primes outside $\mathcal{P}$, for other residue classes, or for $n \neq 7$; [9] had excluded these 1000 primes conditionally on their Conjecture 2.2, which is not used here. $\square$

primes / components	1000 / 32000
T certificates / RHO certificates / T-only components	32000 / 31987 / 13
largest certified T / floor / minimal margin	4164.4897 / 4224 / 59.5103
median certified T	2352.0117
generation	8–20 s per prime; one escalation
local full read-only verification	1000/1000 EXCLUDED, 10 min 42 s wall clock, 4 lanes, 1 BLAS thread each, Apple M4 (10 cores, 24 GB)
negative controls	9/9 rejected (coefficient altered, factor identifier swapped, claimed value altered, one bit of ℓ flipped, duplicated line, header component count altered, header floor altered, header claiming the $n = 2$ floor at $n = 7$, header claiming $33 \cdot 2^2$ on an $n = 2$ file); the untouched $n = 2$ file (floor 68) is accepted

TABLE 6. The certificate family for $\mathcal{P}$ (verification ledger, Appendix F).

5.2. Failure classes. [E] For $\ell \equiv 63, 127 \pmod{128}$ the component minima cluster just above the floor (exact SVP: $\text{ht}/L_7 = 1.009\text{--}1.011$ at $\ell = 10000003199$); the T route closes some of these components and not others. These classes are reported as statistics, not as theorems.

5.3. Limits of the method.

5.3.1. Scaling. Theorem 1.10 and Cor. 1.11 are proved in Appendix E. The lattice half is the group lemma: $[R_n : L_f]$ is a power of the odd prime ℓ , so $L_f \cap 2^{t-1}R_n = 2^{t-1}L_f$ ([F]: `smul_mem_iff_of_coprime`); under hypothesis (i) the filtered lattice is the exact dilate $2^{t-1}\Lambda_f$, and every relative minimum scales by 2^{t-1} . The floor half has two lemmas: the depth- t height floor $\text{ht}(u) \geq L_{n,t}$ for $u \in RE_n^+ \setminus \{\pm 1\}$, $u \equiv 1 \pmod{2^t\mathcal{O}_n}$ (Lemma E.2, Schinzel’s inequality), and the odd-power depth transfer $x^\ell \equiv 1 \iff x \equiv 1 \pmod{2^t\mathcal{O}_n}$ for units x (Lemma E.3; 2 is totally ramified in B_n with residue field $\mathbb{F}_2$, so $(\mathcal{O}_n/2^t\mathcal{O}_n)^\times$ is a 2-group and an odd power map on it is a bijection, [F]: `WeberOddTransfer.pow_odd_eq_one_iff_of_two_pow_card`). The transfer is what allows the floor to be applied to the unit carrying the short vector (r_a , or u_b in the base branch) when only its ℓ -th power u_a is known to be $\equiv 1 \pmod{2^t}$. The analytic half, $(2 + \sqrt{5})^{2^{t-1}} > 2^t + \sqrt{4^t + 1}$ for every $t \geq 2$, is [F]: `ineq_t_general`; it says $L_{n,t}/2^{t-1} < L_n$, so the depth- t contradiction needs a shorter vector than the plain one. Hypothesis (i) is certified for $t = 2, 3, 4$ (Cor. 1.11); for $t \geq 5$ it is open, and we make no claim there.

5.3.2. Class 1. [E] For $\ell \equiv 1 \pmod{128}$ the components have degree 1. For each scalar residue class of a component, the T route may use any lift $a + \ell z$, $z \in R_n$; the search performed used the centered representatives together with an explicit local $\pm\ell$ -coordinate descent, and is not a full nonlinear closest-vector search. At $\ell = 1000000513$ this search over all 64 components found no witness under either floor (per-component minima $5.1 \times 10^4\text{--}2.3 \times 10^5$ against 4224). This is a limitation of the implemented search, not of the T route: the class-1 case above 10^9 and below the Cor. 1.9 threshold is not settled by this method, and no statement about it is made.

5.4. Resources. The costs that matter for trust are the verifier’s and the replayer’s, and only those are recorded as replayable artifacts; generator search time is outside the trusted base, and the generation times quoted in Section 1 and Table 6 are observations ([E]), not replayable artifacts. Table 7 collects the recorded measurements: the clean re-verification of the full family, from the verification ledger of Appendix F (which records the verifier’s SHA-256 and the Sage version alongside every verdict), and the per-profile times of the clean-checkout CI replay on GitHub-hosted runners at the release commit of the version of record (the run identifier is in the release metadata, and the API record of the run is public). No local production machine is part of any claim.

task	scale and arithmetic	wall time	artifacts
<i>Untrusted generation ([E], observational; input to no proof)</i>			
generation	LLL/BKZ + SVP, 1000 primes	8–20 s/prime	witness files
<i>Trusted verification and replay</i>			
family verification	1000 × 32 components, balls 256 bits	642 s	1000 files, 47 MB
CI, Sage profile	Sage gates of the portable replay	873 s	normalized logs
CI, Lean profile	pinned workspace, axiom census	51 s + 100 s fetch	compile logs
C_n certificate	$2 \leq n \leq 9$, balls 500 bits	in Sage profile	20 KB
$p = 3$ certificate	120 intervals, balls 4000 bits	in Sage profile	4.2 MB

TABLE 7. Resource measurements. Generation (top): as reported in Section 1.3 (one prime required an escalation); hardware and thread counts were not recorded, and the values are observations, not inputs to any proof. Family verification: the clean re-verification ledger (Sage 10.8; verifier SHA-256 0f38bb0d...; 1000/1000 EXCLUDED; 63987 accepted certificate lines, 32000 T and 31987 RHO; the ledger also records $\max T = 4164.48\dots$ against $\bar{T}_n = 4224$, margin $59.51\dots$, and median $T = 2352.01\dots$). CI: GitHub-hosted `ubuntu-latest`, tag `v1.0.2`, commit `741dcb48`, run `33292431840`; step durations from the public API record of that run. The measured values are retained as the record of that run; the release run of the tag of record is named in that tag’s release metadata.

6. THE CYCLOTOMIC $\mathbb{Z}_3$ -TOWER

Theorem SH is not tied to $p = 2$. In the cyclotomic $\mathbb{Z}_3$ -tower the saturation input is Horie’s lemma in the form of [12] Lemma 1.3 (the ℓ -th-root statement on the ideal lattice $\ell\mathcal{L}^{-1}$ of $\mathbb{Z}[\zeta_{3^r}]$ acting on the Horie unit), the floor is the relative-norm-one height bound [13] Lemma 2.5(2), applicable because the Horie unit has relative norm 1 (Lemma 6.1), and the covolume of the Horie-unit log lattice is positive at every layer and is computed exactly by a character product (Theorem 6.3). The result is a layer-fixed criterion of the same shape as Theorem A which, at $n = 4$, is sharper than the uniform-in- n bound of Morisawa and Okazaki ([13] Thm. A) in both classes $\ell \equiv \pm 1 \pmod{81}$ with $81 \nmid \ell \mp 1$ (Corollary 6.6).

Throughout, $B_{3,n}$ denotes the n -th layer of the cyclotomic $\mathbb{Z}_3$ -extension of $\mathbb{Q}$ (the real subfield of $\mathbb{Q}(\zeta_{3^{n+1}})$ of degree 3^n), with $B_{3,0} = \mathbb{Q}$ and $h_{3,0} = 1$, so that for $n = 1$ the relative class number $h_{3,1}/h_{3,0}$ is $h_{3,1}$ itself; $E_{3,n}$ is the unit group of $B_{3,n}$, $h_{3,n}$ its class number; $N = 3^n$, $q = 3^{n+1}$, $\zeta = \zeta_q$, σ the generator of $G = \text{Gal}(B_{3,n}/\mathbb{Q})$ induced by $\zeta \mapsto \zeta^4$, $\tau = \sigma^{3^{n-1}}$, $\eta_n = \sin(2(1 + 3^n)\pi/q)/\sin(2\pi/q)$ the Horie unit ([12] §1; one factor at $p = 3$), and H the log map to $\mathbb{R}^N$ over the real embeddings. Throughout the $\mathbb{Z}_3$ -tower, “ $\ell \mid h_{3,n}/h_{3,n-1}$ ” means $v_\ell(h_{3,n}) > v_\ell(h_{3,n-1})$, and “ $\ell \nmid h_{3,n}/h_{3,n-1}$ ” its negation; by Lemma 2.1 with $p = 3$ and $\ell \neq 3$ (the extension $B_{3,n}/B_{3,n-1}$

is cyclic of degree 3) this is $\ker(N)_\ell \neq 0$, and $v_\ell(h_{3,n}) \geq v_\ell(h_{3,n-1})$ always holds, so the quotient is ℓ -integral for every $\ell \neq 3$.

Lemma 6.1 (relative norm of the Horie unit). *$a = 1 + 3^n$ has exact order 3 modulo q and τ acts by $\zeta \mapsto \zeta^a$; $\text{Nr}_{B_{3,n}/B_{3,n-1}}(\eta_n) = 1$. If $\eta_n^{\alpha_\sigma} = \varepsilon^\ell$ with $\alpha_\sigma \in \mathbb{Z}[G]$, $\varepsilon \in E_{3,n}$, ℓ an odd prime, then $\text{Nr}_{B_{3,n}/B_{3,n-1}}(\varepsilon) = 1$. In general, for $\varepsilon \in E_{3,n-1}$ one has $\text{Nr}_{B_{3,n}/B_{3,n-1}}(\varepsilon) = \varepsilon^3$, so a unit of the lower layer with relative norm 1 equals 1; hence a unit of relative norm 1 with $H(\varepsilon) \neq 0$ is not in $E_{3,n-1}$.*

Proof. Let $N = 3^n$, $q = 3N$, $\zeta = \zeta_q$, $a = 1 + N$ and $\tau = \sigma^{3^{n-1}}$, where σ is induced by $\zeta \mapsto \zeta^4$.

(i) *a has exact order 3 modulo q .* $(1 + N)^3 - 1 = 3N + 3N^2 + N^3 = q(1 + N + N^2/3)$ with $3 \mid N^2$ for $n \geq 1$, so $a^3 \equiv 1 \pmod{q}$ (`WeberP3Rel.a_cube_modEq_one`); and $a \not\equiv 1 \pmod{q}$ since $0 < N < q$, so the order is exactly 3 (the only divisors of 3 being 1 and 3).

(ii) *τ acts by $\zeta \mapsto \zeta^a$.* By induction $4^{3^k} = 1 + 3^{k+1} + 3^{k+2}t_k$ with $t_k \in \mathbb{Z}$ (from $(1+u)^3 = 1+3u+3u^2+u^3$, $u = 3^{k+1}(1+3t_{k-1})$), hence $4^{3^{n-1}} \equiv 1+3^n = a \pmod{3^{n+1}}$ (`WeberP3Rel.four_pow_modEq`). Since $G = \text{Gal}(B_{3,n}/\mathbb{Q})$ is cyclic of order N and τ has order 3 by (i), τ generates its unique subgroup of order 3, which is $\text{Gal}(B_{3,n}/B_{3,n-1})$.

(iii) *Relative norm.* For t prime to 3, $\sigma_t : \zeta \mapsto \zeta^t$ sends $\eta_n = (\zeta^a - \zeta^{-a})/(\zeta - \zeta^{-1}) = \sin(2a\pi/q)/\sin(2\pi/q)$ to $\sin(2ta\pi/q)/\sin(2t\pi/q)$. With $s(i) = \sin(2a^i\pi/q)$ ($i = 0, 1, 2, 3$; all nonzero since $q \nmid 2a^i$) this gives $\tau^i \eta_n = s(i+1)/s(i)$ and

$$\text{Nr}_{B_{3,n}/B_{3,n-1}}(\eta_n) = \eta_n \cdot \tau \eta_n \cdot \tau^2 \eta_n = \frac{s(1)s(2)s(3)}{s(0)s(1)s(2)} = \frac{s(3)}{s(0)} = 1,$$

because $a^3 = 1 + mq$ gives $s(3) = \sin(2\pi/q + 2\pi m) = s(0)$ (`WeberP3Rel.sin_telescope_step`, `WeberP3Rel.telescope_three`).

(iv) *Transfer to the saturation root.* Let $\alpha_\sigma \in \mathbb{Z}[G]$ and $\varepsilon \in E_{3,n}$ with $\eta_n^{\alpha_\sigma} = \varepsilon^\ell$, ℓ an odd prime. As G is abelian, Nr commutes with α_σ , so $\text{Nr}(\varepsilon)^\ell = \text{Nr}(\eta_n)^{\alpha_\sigma} = 1$; $\text{Nr}(\varepsilon)$ is a real number ($B_{3,n-1}$ is totally real) and ℓ is odd, hence $\text{Nr}(\varepsilon) = 1$ (`WeberP3Rel.eq_one_of_odd_pow_eq_one`). If moreover $H(\varepsilon) \neq 0$ then $\varepsilon \notin E_{3,n-1}$ by (v) below. Thus every such ε with $H(\varepsilon) \neq 0$ lies in $E_{3,n} \setminus E_{3,n-1}$ and has relative norm 1; the subgroup $U_{3,n} := \{\varepsilon \in E_{3,n} : \text{Nr}_{B_{3,n}/B_{3,n-1}} \varepsilon = 1\}$ contains all ℓ -th roots produced by Horie's lemma ([12] Lemma 1.3).

(v) *Units of the lower layer with relative norm 1.* Let $\varepsilon \in E_{3,n-1}$. The Galois group of $B_{3,n}/B_{3,n-1}$ is $\langle \tau \rangle$, of order 3, and it fixes ε , so

$$\text{Nr}_{B_{3,n}/B_{3,n-1}}(\varepsilon) = \varepsilon \cdot \tau \varepsilon \cdot \tau^2 \varepsilon = \varepsilon^3.$$

Hence $\text{Nr}(\varepsilon) = 1$ gives $\varepsilon^3 = 1$ with ε real, so $\varepsilon = 1$ (`WeberP3Rel.eq_one_of_odd_pow_eq_one` with the odd exponent 3). Consequently a unit $\varepsilon \in U_{3,n}$ with $H(\varepsilon) \neq 0$ cannot lie in $E_{3,n-1}$: it would equal 1 and have $H(\varepsilon) = 0$. $\square$

Lemma 6.2 (discriminant of $\mathbb{Q}(\zeta_{p^r})$). *For a prime p and $r \geq 1$, $|\text{disc } \mathbb{Q}(\zeta_{p^r})| = p^{p^{r-1}(pr-r-1)}$ and $\mathbb{Z}[\zeta_{p^r}]$ is the ring of integers of $\mathbb{Q}(\zeta_{p^r})$ ([20], Chapter 2). In particular $|\text{disc } \mathbb{Q}(\zeta_{3^r})| = 3^{3^{r-1}(2r-1)}$, and this equals $\prod_{b \neq b'} |\xi^b - \xi^{b'}|$ over the $\varphi(3^r)$ primitive 3^r -th roots of unity ξ^b , $b \in (\mathbb{Z}/3^r)^\times$, because the discriminant of the power basis $1, \zeta, \dots, \zeta^{\varphi(3^r)-1}$ is $\text{disc } \Phi_{3^r} = \prod_{b \neq b'} (\xi^b - \xi^{b'})$ and the power basis is an integral basis.*

Theorem 6.3 (spectral rank and exact covolume). *Let $\lambda_j = \log |\sigma^j \eta_n|$ ($j \in \mathbb{Z}/N$) and $\hat{\lambda}_k = \sum_j \lambda_j e^{-2\pi i j k / N}$. Then $\hat{\lambda}_k = 0$ if $3 \mid k$ and $\hat{\lambda}_k \neq 0$ if $3 \nmid k$. Consequently, for every $n \geq 1$ and $1 \leq r \leq n$, with $c = 2 \cdot 3^{r-1}$, the vectors $H(\sigma^{3^{n-r}i} \eta_n)$, $0 \leq i < c$, are linearly independent, and their covolume $D_r^{(n)}$ satisfies*

$$(D_r^{(n)})^2 = 3^{3^{r-1}(2r-1)} \prod_{b \in (\mathbb{Z}/3^r)^\times} W_{n,r}(b), \quad W_{n,r}(b) = \frac{1}{N} \sum_{k \equiv b \pmod{3^r}} |\hat{\lambda}_k|^2 > 0.$$

Proof. Fix $n \geq 1$, $N = 3^n$, $\omega = e^{2\pi i / N}$; for $j \in \mathbb{Z}/N$ let $\lambda_j = \log |\sigma^j \eta_n|$ and $\hat{\lambda}_k = \sum_{j \in \mathbb{Z}/N} \lambda_j \omega^{-jk}$ ($k \in \mathbb{Z}/N$).

Step 1: $\hat{\lambda}_k = 0$ for $3 \mid k$. By Lemma 6.1(iii), $\lambda_j + \lambda_{j+N/3} + \lambda_{j+2N/3} = \log |\sigma^j \text{Nr}_{B_{3,n}/B_{3,n-1}} \eta_n| = 0$ for every j ($\tau = \sigma^{N/3}$). For $k = 3k'$ one has $\omega^{-(j+N/3)k} = \omega^{-jk}$, so grouping $\mathbb{Z}/N$ into the $N/3$ cosets of $\{0, N/3, 2N/3\}$ gives $\hat{\lambda}_k = \sum_{j < N/3} \omega^{-jk} (\lambda_j + \lambda_{j+N/3} + \lambda_{j+2N/3}) = 0$ (`WeberP3Rel.dft_vanish_of_relnorm`).

Step 2: $\hat{\lambda}_k \neq 0$ for $3 \nmid k$. Since $|\sin(2\pi m/q)| = \frac{1}{2}|1 - \zeta_q^{2m}|$, $\lambda_j = g(2a \cdot 4^j) - g(2 \cdot 4^j)$ with $g(m) = \log |1 - \zeta_q^m|$ on $(\mathbb{Z}/q)^\times = \{\pm 1\} \times \langle 4 \rangle$ ($q = 3N$, $a = 1 + N$, and $a \equiv 4^{N/3} \pmod{q}$ by Lemma 6.1(ii), `WeberP3Rel.four_pow_modEq`; the arguments of g , and below of χ_k , are residue classes modulo q). As $g(-m) = g(m)$ and $-2 \in \langle 4 \rangle$, write $-2 \equiv 4^{j_0} \pmod{q}$; then $\lambda_j = \Gamma(j + j_0 + N/3) - \Gamma(j + j_0)$ with $\Gamma(j) = g(4^j)$, hence $\hat{\lambda}_k = \omega^{j_0 k} (\omega^{Nk/3} - 1) \hat{\Gamma}_k$ (shift rule of the twisted sum on $\mathbb{Z}/N$: `WeberP3Rel.twisted_sum_shift`, `WeberP3Rel.twisted_sum_shift_diff`) and $\omega^{Nk/3} = e^{2\pi i k / 3} \neq 1$ (`WeberP3Rel.pow_third_eq_one_iff`); so $\hat{\lambda}_k \neq 0$ as soon as $\hat{\Gamma}_k \neq 0$ (`WeberP3Rel.twisted_sum_shift_diff_ne_zero`). Let χ_k be the character of $(\mathbb{Z}/q)^\times = \{\pm 1\} \times \langle 4 \rangle$ trivial on -1 with $\chi_k(4^j) = \omega^{-jk}$ (well defined since 4 has order N in $(\mathbb{Z}/q)^\times$ and $\omega^N = 1$); it is even. *Conductor.* Every proper divisor of $q = 3^{n+1}$ divides 3^n , so χ_k is imprimitive iff it factors through the reduction $(\mathbb{Z}/q)^\times \rightarrow (\mathbb{Z}/3^n)^\times$, i.e. iff it is trivial on the kernel of that reduction. The kernel is $\{t \in (\mathbb{Z}/q)^\times : t \equiv 1 \pmod{3^n}\} = \{1, 1 + 3^n, 1 + 2 \cdot 3^n\} = \{1, a, a^2\}$ (order 3; $a = 1 + N$ has exact order 3 by Lemma 6.1(i), and $a^2 \equiv 1 + 2N \pmod{q}$ as $N^2 \equiv 0 \pmod{q}$). Since $a \equiv 4^{N/3} \pmod{q}$ (Lemma 6.1(ii), `WeberP3Rel.four_pow_modEq`) and χ_k is a function of the class modulo q , $\chi_k(a) = \omega^{-Nk/3} = e^{-2\pi i k / 3} \neq 1$ for $3 \nmid k$. So χ_k is not trivial on the kernel: it is primitive of conductor exactly $q = 3^{n+1}$. *Conventions.* Throughout,

$\omega = e^{2\pi i/N}$ and $\hat{\lambda}_k = \sum_j \lambda_j \omega^{-jk}$ (minus sign in the exponent; the same sign is used in the certificate script and in the read-only verifier, and the Lean statements of `WeberP3Rel` are proved for an arbitrary ω with $\omega^N = 1$, so they cover either sign); only $|\hat{\Gamma}_k|$ enters below, through the absolute-value form of Washington's formula (Lemma D.4), so no Gauss-sum normalisation is used. Then $\hat{\Gamma}_k = \sum_{m \in \langle 4 \rangle} \chi_k(m) g(m) = \frac{1}{2} \sum_{m \in (\mathbb{Z}/q)^\times} \chi_k(m) \log |1 - \zeta_q^m|$, whose absolute value is $\frac{\sqrt{q}}{2} |L(1, \chi_k)|$ by Lemma D.4 (Washington's formula, [20], Chapter 4), and $L(1, \chi_k) \neq 0$ by Dirichlet's theorem. So $\hat{\lambda}_k \neq 0$.

Step 3: rank c for every n and $1 \leq r \leq n$. Put $h = 3^{n-r}$, $c = 2 \cdot 3^{r-1} = \varphi(3^r)$ and $\xi = \omega^h$, a primitive 3^r -th root of unity. For $a_0, \dots, a_{c-1} \in \mathbb{R}$ the vector $v = \sum_{i < c} a_i H(\sigma^{hi} \eta_n)$ has coordinates $v_j = \sum_i a_i \lambda_{j+hi}$ and transform $\hat{v}_k = A(\omega^{hk}) \hat{\lambda}_k$, $A(x) = \sum_i a_i x^i$. If $v = 0$ then $A(\xi^k) = 0$ for all $3 \nmid k$ by Step 2; as k runs over $(\mathbb{Z}/3^r)^\times$ the ξ^k are the c primitive 3^r -th roots of unity, and $\deg A < c$ forces $A = 0$ (`WeberP3Rel.eq_zero_of_vanish_on_primitiveRoots`, `WeberP3Rel.totient_three_pow`). Hence the c vectors $H(\sigma^{hi} \eta_n)$ are linearly independent: (Rank) holds and $D_r^{(n)} > 0$.

Step 4: exact covolume. With Parseval's identity $\langle u, v \rangle = N^{-1} \sum_k \hat{u}_k \overline{\hat{v}_k}$ the Gram matrix of the c vectors is

$$G_{ii'} = \frac{1}{N} \sum_k |\hat{\lambda}_k|^2 \omega^{hk(i-i')} = \sum_{b \in (\mathbb{Z}/3^r)^\times} W_{n,r}(b) \xi^{b(i-i')},$$

where $W_{n,r}(b) = \frac{1}{N} \sum_{k \equiv b \pmod{3^r}} |\hat{\lambda}_k|^2 > 0$, the terms $3 \mid k$ being 0 by Step 1 and the others positive by Step 2. *Normalisation.* The left side $G_{ii'} = \sum_{j \in \mathbb{Z}/N} \lambda_{j+hi} \lambda_{j+hi'}$ is the Gram matrix of the log vectors themselves (the Gram route of the certificate computes it directly from the λ_j), and the right side is Parseval's identity for the unnormalised transform $\hat{u}_k = \sum_j u_j \omega^{-jk}$, whose inverse carries the factor $1/N$; this is the only place where $1/N$ enters, and it is the $1/N$ in the definition of $W_{n,r}(b)$ (the character-product route computes the right side from the $\hat{\lambda}_k$). Thus $G = V \operatorname{diag}(W) V^*$ with $V = (\xi^{bi})_{i < c, b \in (\mathbb{Z}/3^r)^\times}$, and $|\det V|^2 = \prod_{b \neq b'} |\xi^b - \xi^{b'}| = |\operatorname{disc} \Phi_{3^r}| = |\operatorname{disc} \mathbb{Q}(\zeta_{3^r})| = 3^{3^{r-1}(2r-1)}$ (Vandermonde over the c primitive 3^r -th roots of unity; the discriminant is Lemma 6.2). Therefore $(D_r^{(n)})^2 = \det G = 3^{3^{r-1}(2r-1)} \prod_b W_{n,r}(b)$, which is the second route of the certificate [P3-CERT]: the two routes are independent linear-algebraic evaluations from the same log profile (λ_j) (Gram determinant; character product), and they agree to the ball radius in all 15 rows. $\square$

Lemma 6.4 (relative-norm-one height floor). *If $\varepsilon \in E_{3,n} \setminus E_{3,n-1}$ and $\operatorname{Nr}_{B_{3,n}/B_{3,n-1}}(\varepsilon) = 1$, then $\operatorname{ht}(\varepsilon) \geq L_{3,n}^{\operatorname{rel}} := \sqrt{3^n} \log \frac{3^{(3^n-1)/(2 \cdot 3^n)} + \sqrt{3^{(3^n-1)/3^n} + 4}}{2}$, where ht is the Euclidean norm of the log vector ([13] Def. 2.2). ([13] Lemma 2.5(2): Lemma 2.3 there, the Mahler-measure bound with*

$C = |\mathrm{Nr}_{B_{3,n}/\mathbb{Q}}(\varepsilon^2 - 1)|$, together with Lemma 2.4(2), $C \geq 3^{(3^n-1)/2}$, which is [12] Lemma 9.1.)

Theorem 6.5 (P3). *Let $n \geq 1$, $\ell \neq 3$ an odd prime, $f \in \{1, 2\}$ the order of ℓ modulo 3, $3^s \parallel \ell^2 - 1$, $r = \min(n, s)$, $c = 2 \cdot 3^{r-1}$, and $L_{3,n}^{\mathrm{rel}}$ as in Lemma 6.4. Let $D_r^{(n)} > 0$ be the covolume of Theorem 6.3 and put*

$$C_{n,r}^{(3)} = (2/\pi)^{c/2} \Gamma(2 + c/2) D_r^{(n)} / (L_{3,n}^{\mathrm{rel}})^c.$$

If $\ell^f > C_{n,r}^{(3)}$ then $\ell \nmid h_{3,n}/h_{3,n-1}$.

Proof. The conclusion. “ $\ell \nmid h_{3,n}/h_{3,n-1}$ ” is read as $v_\ell(h_{3,n}) = v_\ell(h_{3,n-1})$, equivalently $\ker(N)_\ell = 0$ for the norm N of $B_{3,n}/B_{3,n-1}$ (Lemma 2.1 with $p = 3$, $\ell \neq 3$). *The prime.* Since ℓ is odd and $\ell \neq 3$, the two places where the oddness of ℓ enters are available: Lemma 6.1(iv) (the transfer of relative norm 1 to the ℓ -th root, where $\mathrm{Nr}(\varepsilon)^\ell = 1$ with $\mathrm{Nr}(\varepsilon)$ real forces $\mathrm{Nr}(\varepsilon) = 1$, `WeberP3Rel.eq_one_of_odd_pow_eq_one`) and the hypothesis of Theorem 1.1, which is stated for an odd prime. Nothing below is claimed for $\ell = 2$.

The data. Let $p = 3$, $N = 3^n$, $q = 3^{n+1}$, $\zeta = e^{2\pi i/q}$, and let σ be the generator $\zeta \mapsto \zeta^4$ of $\mathrm{Gal}(\mathbb{Q}(\zeta)/\mathbb{Q}(\zeta^{3^n})) \cong \mathbb{Z}/3^n$, which restricts to a generator of $G = \mathrm{Gal}(B_{3,n}/\mathbb{Q})$. The Horie unit is $\eta_n = \sin(2(1 + 3^n)\pi/q) / \sin(2\pi/q)$ ([12] §1; at $p = 3$ the product defining η_n has one factor), a unit of $B_{3,n}$ of relative norm 1 (Lemma 6.1). For $\alpha = \sum_{i < c_n} a_i \zeta_{3^n}^i \in \mathbb{Z}[\zeta_{3^n}]$ ($c_n = 2 \cdot 3^{n-1}$, $\zeta_{3^n} = \zeta^3$) put $\alpha_\sigma = \sum_{i < c_n} a_i \sigma^i$ and $u_\alpha = \eta_n^{\alpha_\sigma}$; for $\alpha \in \mathbb{Z}[\zeta_{3^r}] \subseteq \mathbb{Z}[\zeta_{3^n}]$ (via $\zeta_{3^r} = \zeta_{3^n}^{3^{n-r}}$) the exponents of α_σ are the multiples $3^{n-r}j$, $j < c$. The map $\alpha \mapsto H(u_\alpha)$ is additive in the coordinates a_j , and its image on $\mathbb{Z}[\zeta_{3^r}]$ is the lattice spanned by $H(\sigma^{3^{n-r}j} \eta_n)$, $j < c$, whose covolume $D_r^{(n)}$ is positive by Theorem 6.3: this is (Rank) of Theorem 1.1, for every n and r .

Saturation (the input of [12] Lemma 1.3). Since $r \leq s$, the field $F = \mathbb{Q}(\zeta_{3^r})$ contains the decomposition field of ℓ in $\mathbb{Q}(\zeta_{3^n})$ (for $f = 1$ the decomposition field is $\mathbb{Q}(\zeta_{3^{\min(n,s)}})$ itself; for $f = 2$ it is its maximal real subfield). If $\ell \mid h_{3,n}/h_{3,n-1}$, [12] Lemma 1.3 (attributed there to [6], which relays [3]; the general form with proof is [5] Prop. 1) gives a prime $\mathcal{L}$ of F above ℓ such that $u_\alpha = \eta_n^{\alpha_\sigma}$ is an ℓ -th power in $E_{3,n}$ for every $\alpha \in \ell\mathcal{L}^{-1}$. The residue degree of $\mathcal{L}$ is f , so $\ell\mathcal{L}^{-1}$ has index ℓ^{c-f} in $\mathbb{Z}[\zeta_{3^r}]$: this is (Index) with $d = f$. Let $U = U_{3,n} = \{\varepsilon \in E_{3,n} : \mathrm{Nr}_{B_{3,n}/B_{3,n-1}} \varepsilon = 1\}$. (Car) for $\alpha \in \ell\mathcal{L}^{-1} \setminus \ell\mathbb{Z}[\zeta_{3^r}]$: the ℓ -th root ε has $\ell H(\varepsilon) = H(u_\alpha)$ (`WeberSH.sat_root_log`) and lies in $U_{3,n}$ by Lemma 6.1(iv). (Car) for $\alpha = \ell\beta$: u_β itself carries $H(u_\alpha) = \ell H(u_\beta)$ (`WeberSH.base_branch`), and $u_\beta \in U_{3,n}$ because $\mathrm{Nr}(\eta_n^{\beta_\sigma}) = \mathrm{Nr}(\eta_n)^{\beta_\sigma} = 1$ (Lemma 6.1(iii)).

The floor. Let $\varepsilon \in U_{3,n}$ with $H(\varepsilon) \neq 0$. By definition of $U_{3,n}$, $\mathrm{Nr}_{B_{3,n}/B_{3,n-1}}(\varepsilon) = 1$; by Lemma 6.1(v) (a unit of $E_{3,n-1}$ with relative norm 1 equals 1, and $H(1) = 0$), $\varepsilon \notin E_{3,n-1}$. Hence $\varepsilon \in E_{3,n} \setminus E_{3,n-1}$ with relative norm 1, and Lemma 6.4 gives $\mathrm{ht}(\varepsilon) \geq L_{3,n}^{\mathrm{rel}}$ directly in the

Euclidean height (the height of [13] Def. 2.2 is the ℓ^2 -norm of the log vector; no Cauchy–Schwarz step is needed). This is (Floor) for $U = U_{3,n}$ with $L_0 = L_{3,n}^{\text{rel}}$.

Conclusion. Theorem 1.1 (`WeberSH.theoremSH_contra`; `WeberP3.theoremP3_core` is the same statement with the $p = 3$ names) with $K = B_{3,n}$, $m = c$, the lattice $\ell\mathcal{L}^{-1}$, $L_0 = L_{3,n}^{\text{rel}}$ and Blichfeldt’s theorem ([1] Thm. II as [13] Thm. 2.7) on $\Lambda = \ell^{-1}H(u_{\ell\mathcal{L}^{-1}})$, of rank c and covolume $D_r^{(n)}/\ell^f$ in its span, yields $\ell^f \leq (2/\pi)^{c/2}\Gamma(2 + c/2)D_r^{(n)}/(L_{3,n}^{\text{rel}})^c = C_{n,r}^{(3)}$. Contrapositively, $\ell^f > C_{n,r}^{(3)}$ implies $\ell \nmid h_{3,n}/h_{3,n-1}$.

Remark (what changes relative to [12], [13]). The saturation input is that of [12]; the floor is that of [13] Lemma 2.5(2), applicable because $\text{Nr } \eta_n = 1$ transfers to the saturation root (Lemma 6.1); the change is the convex body: [12] §4–5 and [13] §4 apply Minkowski’s or Blichfeldt’s theorem to an ideal lattice in coordinates and bound the height of δ_n^α through the Mahler measure, which produces the factors $c!$ resp. $((c+2)/2)! (\sqrt{2\pi}/(3^{3/4} \log(\dots)))^c$ in $G_1(3, s, f)^f$ and $G(3, s, f)^f$; here Blichfeldt’s theorem acts on the log lattice with its exact covolume $D_r^{(n)}$ (Theorem 6.3), and the floor enters through $(L_{3,n}^{\text{rel}})^c$ with $L_{3,n}^{\text{rel}}$ growing like $\sqrt{3^n}$. $\square$

Corollary 6.6 ($n = 4$). *If $\ell \equiv 1 \pmod{81}$ and $\ell > 1.0728 \times 10^{33}$, or $\ell \equiv -1 \pmod{81}$ and $\ell > 3.2753 \times 10^{16}$, then $\ell \nmid h_{3,4}/h_{3,3}$. In the classes $81 \parallel \ell \mp 1$ these layer-fixed thresholds are below the uniform-in- n bound of Morisawa and Okazaki, [13] Thm. A (3.7111×10^{37} , resp. 6.0919×10^{18}), by the factors 3.4594×10^4 , resp. 1.8599×10^2 .*

Proof. At $n = 4$ we have $N = 81$, $s \geq 4$ means $81 \mid \ell^2 - 1$, i.e. $\ell \equiv \pm 1 \pmod{81}$, so $r = 4$, $c = 54$, and $f = 1$ for $\ell \equiv 1$, $f = 2$ for $\ell \equiv -1 \pmod{81}$; $L_{3,4}^{\text{rel}} = 9 \log \frac{3^{40/81} + \sqrt{3^{80/81} + 4}}{2} = 7.0107178918636975 \dots$. The certificate [P3-CERT] is the outward hull, widened to twice its half-width, of two independent 4000-bit ball enclosures of every quantity below: the producer’s (the 81 logarithms $\log |\sigma^t \eta_4|$ are balls, the 54×54 Gram matrix and its determinant are ball operations, and independently the discrete Fourier transform of the log vector and the character product of Theorem 6.3 Step 4; the two enclosures of $D_4^{(4)}$ have radius below 10^{-1000} and their quotient encloses 1) and the read-only replay’s [P3-REPLAY], which rebuilds the same quantities from the definitions by Gram–Schmidt orthogonalisation and a table of powers of $e^{-2\pi i/N}$; the certificate is generated by [P3-GEN], and the replay requires every certified interval to contain its own recomputation (coverage checker [P3-COVER]). It encloses

$$C_{4,4}^{(3)} = \frac{(2/\pi)^{27} \Gamma(29) D_4^{(4)}}{(L_{3,4}^{\text{rel}})^{54}} \in [1.072749779 \times 10^{33}, 1.072749780 \times 10^{33}],$$

$$\sqrt{C_{4,4}^{(3)}} \in [3.275285910 \times 10^{16}, 3.275285911 \times 10^{16}].$$

n	r	c	$C_{n,r}^{(3)}$ (up)	$\sqrt{C_{n,r}^{(3)}}$ (up)	$G(3, r, 1)$ [13]	$G(3, r, 2)$ [13]	gain $f=1$	gain $f=2$
2	2	6	4.8939×10^1	6.9957×10^0	1.8992×10^2	1.3781×10^1	3.88×10^0	1.96×10^0
3	3	18	5.4438×10^7	7.3782×10^3	1.7983×10^9	4.2407×10^4	3.30×10^1	5.74×10^0
4	1	2	3.7148×10^0	1.9274×10^0	3.9855×10^0	1.9963×10^0	1.07×10^0	1.03×10^0
4	2	6	1.5360×10^2	1.2394×10^1	1.8992×10^2	1.3781×10^1	1.23×10^0	1.11×10^0
4	3	18	8.8915×10^8	2.9819×10^4	1.7983×10^9	4.2407×10^4	2.02×10^0	1.42×10^0
4	4	54	1.0728×10^{33}	3.2753×10^{16}	3.7111×10^{37}	6.0919×10^{18}	3.45×10^4	1.85×10^2
5	4	54	6.1141×10^{36}	2.4727×10^{18}	3.7111×10^{37}	6.0919×10^{18}	6.06×10^0	2.46×10^0
5	5	162	1.7382×10^{133}	4.1691×10^{66}	8.5728×10^{146}	2.9279×10^{73}	4.93×10^{13}	7.02×10^6

TABLE 8. The layer-fixed constant $C_{n,r}^{(3)}$ of Theorem 6.5 and its square root (thresholds for $f = 1, 2$; [C]: the $p = 3$ certificate of Appendix F, ball arithmetic at 4000 bits, every row an enclosure of radius $< 10^{-800}$ by two independent linear-algebraic routes from the same log profile (Gram determinant; character product), stored as exact dyadic outward endpoints and replayed from the definitions by its read-only verifier; upper endpoints, rounded up) against the uniform-in- n bound of Morisawa and Okazaki, [13] Thm. A, evaluated at $s = r$ (lower endpoints, rounded down), and the gains $G(3, r, 1)/C_{n,r}^{(3)}$, $G(3, r, 2)/\sqrt{C_{n,r}^{(3)}}$ (rounded down). The present constant is below G in both classes in every row $1 \leq r \leq n \leq 5$ (all fifteen rows are in the certificate). For $s > r = n$ the comparator is $G(3, s, f)$, which grows with $c = 2 \cdot 3^{s-1}$, while $C_{n,n}^{(3)}$ is fixed. The comparison is between a layer-fixed constant and a uniform-in- n bound: $G(3, s, f)$ holds at every layer $n \geq s$, while $C_{n,r}^{(3)}$ is a bound at the single layer n . For the historical record, [12] Thm. 0.3 needs $\ell^f > 2^{c/2}c!$ (3.098×10^{79} at $s = 4$) and [12] Thm. A $\ell^f > G_1(3, s, f)^f$ (7.63×10^{101}).

Hence $\ell \equiv 1 \pmod{81}$ and $\ell > 1.0728 \times 10^{33}$ give $\ell^f = \ell > C_{4,4}^{(3)}$, and $\ell \equiv -1 \pmod{81}$ and $\ell > 3.2753 \times 10^{16}$ give $\ell^f = \ell^2 > C_{4,4}^{(3)}$; Theorem 6.5 applies. The same certificate gives the rows $n \leq 5$ of the comparison table in the $\mathbb{Z}_3$ section of the paper. For the comparison: the uniform-in- n bound of Morisawa and Okazaki, [13] Thm. A, which for $p = 3$, $s = 4$ reads $\ell > G(3, 4, f)$ with $G(3, s, f) = ((\sqrt{2\pi}/(3^{3/4} \log \frac{3^{40/81} + \sqrt{3^{80/81} + 4}}{2}))^c ((c+2)/2)!)^{1/f}$, $c = 2 \cdot 3^{s-1}$, i.e. $G(3, 4, 1) \geq 3.7111 \times 10^{37}$ and $G(3, 4, 2) \geq 6.0919 \times 10^{18}$ (the same certificate encloses G ; its value $G(3, 3, 2) = 42407.49\dots$ agrees with the table of [13] Example 1.6, “ 4.3×10^4 ”). In the class $\ell \equiv 1 \pmod{81}$ with $81 \parallel \ell - 1$ (the set $D(3, 4, 1)$ of [13]) the present threshold is smaller by the factor $G(3, 4, 1)/C_{4,4}^{(3)} \geq 3.4594 \times 10^4$, and in $D(3, 4, 2)$ ($\ell \equiv -1 \pmod{81}$),

$81 \parallel \ell+1$) by $G(3, 4, 2)/\sqrt{C_{4,4}^{(3)}} \geq 1.8599 \times 10^2$. The comparison convention is that of the paper’s comparison section (layer-fixed versus uniform in n): [13] is uniform in n (and its constant for $s \geq 5$ is $G(3, s, f)$, larger), the present bound is for the layer $n = 4$ only; [12] Thm. 0.3 ($2^{c/2}c! = 3.098 \times 10^{79}$ at $s = 4$) and Thm. A ($G_1(3, 4, f)^f = 7.63 \times 10^{101}$) are listed in that table for the record, not as the comparator. $\square$

The layer $n = 4$ is the lowest whose class number is not known unconditionally ($h_{3,n} = 1$ for $n \leq 3$ by [12] §0 and its references; $n = 4$ under the generalized Riemann hypothesis, van der Linden). Nothing is claimed for $h_{3,4}$ itself, only for its relative part at primes in the two classes; for $\ell \equiv \pm 1 \pmod{243}$ the same thresholds apply at the layer $n = 4$ (the theorem uses $r = \min(n, s) = 4$), where the uniform comparator is $G(3, s, f)$ with $s \geq 5$.

7. REPRODUCIBILITY AND TRUST

label	meaning	what it does not cover
F	Lean 4 kernel, std-3 axioms, no sorry , no native_decide	the informal reading of the statement
C	certificate accepted by a read-only checker; checker specification stated	the checker’s implementation (not Lean-verified)
L	external literature result used as an explicit hypothesis (published or preprint; the status and verbatim location on record)	its proof
M	proved in this manuscript and in the Blueprint (single-source proofs /); not machine-checked	kernel or certificate confirmation
E	experiment; used in no theorem	anything

TABLE 9. Trust labels. Composite labels (e.g. “L-relative, F-core”, “C+L, F-core”) name the label of the core and of the inputs separately; the correspondence table and the Blueprint map of the repository (Appendix F) assign them claim by claim.

7.1. Trust labels and correspondence. Every claim carries one of the labels F (Lean 4, toolchain v4.31.0-rc1, Mathlib commit d568c8c0; for every declaration the axiom census returned by `#print axioms` lies within the standard three, `propext`, `Classical.choice` and `Quot.sound`; no **sorry**, no **native_decide**), C (certificate plus read-only checker), L (external literature result used as an explicit hypothesis), M (proved in the manuscript and in the Blueprint, not machine-checked), E (experiment, used in no theorem),

or a composite of these. The correspondence between statements, Lean declarations and certificates is the file `CORRESPONDENCE.csv`; the Lean files, certificates, scripts and logs are indexed in Appendix F. All designated finite computations and selected discrete cores are kernel- or certificate-checked; the remaining arguments (Lemmas B, C (pullback bookkeeping), D, E', Prop. F, Prop. D, the old/new decomposition, Lemma E.3 outside its group core) are M; external literature inputs are isolated explicitly (Table 4).

7.2. Replay and continuous integration. The repository ships a portable replay script (manifest check, cross-CAS certificates for Prop. D and C_7 , the 1000 family certificates, the negative controls, the Lean compilations with axiom census, blueprint and paper builds) and a Dockerfile with a CI workflow for Sage 10.8 and the pinned Lean workspace (the package is replayed from a clean checkout on GitHub-hosted public runners on every push; each tagged release archives its version-specific CI attestation and normalized verification logs as immutable release assets, and the corresponding run identifier and commit SHA are recorded in the release metadata, never written back into the source tree); the release manifest lists every certificate by SHA-256. The acceptance criterion for the package is the replay itself: a clean checkout, the pinned toolchains, and every gate green. Code, certificates and logs are archived on Zenodo; the concept and version DOIs, the tag, the commit SHA and the CI run identifier of the version of record are stated in the release metadata of the corresponding tag, never inside the source tree (a document cannot certify the tree it is part of).

7.3. Relation to prior computations. The prior computations most directly compared with the present work are of three kinds. The unconditional exclusions below 10^9 rest on the sweep of Fukuda and Komatsu [2]; Miller's determination of h_6 and, under the generalized Riemann hypothesis, of h_7 [10] rest on class-number computations whose last step at $n = 7$ consumes the Fukuda–Komatsu exclusion; the exclusions of Kashio and Yoshizaki at $n = 7$ above 10^9 [9] rest on a search whose success is interpreted through their Conjecture 2.2. The package here is organised so that no generator or search run needs to be trusted: every claimed exclusion is accompanied by a certificate that the read-only verifier accepts; the verifier's specification is part of the soundness theorem (Section 3); the search that found the witnesses is outside the trusted base; and the whole package is replayed by continuous integration from a clean checkout. The mathematical scope of the comparison — which primes, which classes, which layers — is the subject of Section 1.3 and Tables 1 and 2; this subsection concerns only the form of the evidence. For the thousand primes of Section 5 the two records overlap exactly: the exclusion that [9] obtain conditionally is obtained here unconditionally, and the certificate is the difference.

7.4. Use of generative AI. Generative AI systems, including Anthropic Claude and OpenAI GPT models, were used throughout this project to assist with the exploration and comparison of proof strategies, tactic-level search and code drafting for the Lean formalization, drafting and revision of the manuscript, and adversarial review of successive drafts. The author made all final mathematical and editorial decisions, checked every definition, statement, proof, and citation included in the manuscript and Blueprint, and assumes sole responsibility for the entire content. As described in this section, designated finite computations and selected discrete proof components were additionally checked by the Lean kernel or by frozen certificate verifiers; these checks do not rely on accepting AI-generated mathematical assertions.

8. OUTLOOK

(1) The class-1 components need a mechanism beyond the implemented T-route search. (2) A coupling between the coset minimum and the height of a deep representative would reopen the $n = 7$ question. (3) Any improvement of the Blichfeldt constant improves C_n at every layer (Section 4). (4) Whether $J_t = 2^{t-1}R_n$ for $t \geq 5$ (Theorem 1.10(i)) is open.

APPENDIX A. THE COMPONENTWISE SATURATION LEMMAS

Lemma B.

Proof. $x^m + 1 = \Phi_{2^n}(x)$ (the 2^n -th cyclotomic polynomial: $\Phi_{2^{k+1}}(x) = \Phi_2(x^{2^k}) = x^{2^k} + 1$); every root ζ has exact order 2^n , and $x^m + 1$ is separable over $\mathbb{F}_\ell$ because $\ell \nmid 2^n$. Over $\mathbb{F}_\ell$ (ℓ odd, unramified) the Frobenius $\zeta \mapsto \zeta^\ell$ permutes the roots, and the degree of the factor through ζ is the least k with $\zeta^{\ell^k} = \zeta$, i.e. $\ell^k \equiv 1 \pmod{2^n}$, the same for every root. (Lean: `WeberLemmaB.factor_degree` and `WeberLemmaB.factor_degree_eq_std-3`, using Mathlib's `Polynomial.natDegree_of_dvd_cyclotomic_of_irreducible` and `WeberLemmaB.X_pow_add_one_eq_cyclotomic`: $x^{2^{n-1}} + 1 = \Phi_{2^n}$.) At $n = 7$, $(\mathbb{Z}/128)^\times$ has exponent 32; $\ell \equiv 1$ has order 1; $63^2 = 3969 = 31 \cdot 128 + 1$, $65^2 = 4225 = 33 \cdot 128 + 1$, $127 \equiv -1$ have order 2. $\square$

Lemma C.

Proof. Since $x^m + 1$ is separable mod ℓ , $\mathbb{F}_\ell[x]/(x^m + 1) = \prod_g \mathbb{F}_\ell[x]/(g)$ over the irreducible factors. $w = (x^m + 1)/f$ vanishes in every g -component with $g \neq f$ and is a unit in the f -component, so $M_f = 0 \times \cdots \times \mathbb{F}_\ell[x]/(f) \times \cdots \times 0$ has dimension d_f , and $R_n/L_f \cong (\mathbb{F}_\ell[x]/(x^m + 1))/M_f$ has dimension $m - d_f$. The Lean statement `component_card_quotient` is the dimension count for the quotient V/W of $\mathbb{F}_\ell$ -spaces; the pullback identification $R_n/L_f \cong V/W$ along π is the bookkeeping just described and is not formalised. $\square$

Lemma D.

Proof. (i) If $H_na = 0$, every real embedding of u_a has absolute value 1; since B_n is totally real, every conjugate of u_a lies in $\{\pm 1\}$, so the minimal polynomial of u_a over $\mathbb{Q}$, being irreducible with all its roots in $\{\pm 1\}$, is $x - 1$ or $x + 1$, and $u_a = \pm 1$. (This is the totally real case of Kronecker's theorem—an algebraic integer all of whose conjugates have absolute value 1 is a root of unity; the general theorem is not needed.) freeness of $A_n/\{\pm 1\}$ ([9] §4) forces $a = 0$. (ii) $H_n(\mathbb{Z}^m)$ is the image of $A_n \subseteq E_n$ under the logarithmic embedding, whose image of E_n is discrete (Dirichlet); a subgroup of a discrete group is discrete. (iii) A discrete torsion-free subgroup of $\mathbb{R}^{2m}$ is free on $\mathbb{R}$ -linearly independent generators; by (i) its $\mathbb{Z}$ -rank is m , so its $\mathbb{R}$ -rank is m . Antiperiodicity $\lambda_{j+m} = -\lambda_j$ gives $(H_na)_{j+m} = -(H_na)_j$, so $H_n^\top H_n = 2W_n^\top W_n$ and $D_n = \sqrt{\det H_n^\top H_n} = 2^{m/2} |\det W_n|$. For Λ_f : L_f has index ℓ^{m-d_f} in $\mathbb{Z}^m$ (Lemma C), so $H_n(L_f)$ has covolume $D_n \ell^{m-d_f}$, and scaling by ℓ^{-1} divides the covolume of a rank- m lattice by ℓ^m . $\square$

Lemma E and Lemma E'. Proved in Section 2.3.

Proposition F (factorwise form). Proved in Section 2.6 (Prop. 2.10); Lemma B then identifies d_{f_0} with $d_n(\ell)$.

APPENDIX B. OLD AND NEW COMPONENTS IN CYCLIC p -TOWERS

Proof of Lemma 2.1. Throughout, $\text{Cl}(\cdot)$ is the ideal class group (finite abelian), $\text{Cl}(\cdot)_\ell$ its ℓ -primary part (the subgroup of elements of ℓ -power order), $\iota : \text{Cl}(K) \rightarrow \text{Cl}(L)$ the map induced by extension of ideals $\mathfrak{a} \mapsto \mathfrak{a}\mathcal{O}_L$, and $N : \text{Cl}(L) \rightarrow \text{Cl}(K)$ the map induced by the ideal norm $N_{L/K}$. Both are group homomorphisms; both restrict to the ℓ -parts because a homomorphism of finite abelian groups maps elements of ℓ -power order to elements of ℓ -power order.

Step 1: $N \circ \iota$ is multiplication by $[L : K]$. For an ideal $\mathfrak{a}$ of K , $N_{L/K}(\mathfrak{a}\mathcal{O}_L) = \mathfrak{a}^{[L:K]}$ (transitivity of the norm: the norm of an extended ideal is the $[L : K]$ -th power). Hence $N \circ \iota = [L : K] = p^k$ on $\text{Cl}(K)$.

Step 2: injectivity on the ℓ -part. Multiplication by p^k is a bijection of the finite abelian ℓ -group $\text{Cl}(K)_\ell$ because $\gcd(p^k, |\text{Cl}(K)_\ell|) = 1$ ($\ell \neq p$); an inverse is multiplication by an integer u with $up^k \equiv 1$ modulo the exponent of $\text{Cl}(K)_\ell$. Since $N \circ \iota$ is bijective on $\text{Cl}(K)_\ell$, ι is injective there, and $\iota(\text{Cl}(K)_\ell) \cap \ker N = 0$: an element ιy with $y \in \text{Cl}(K)_\ell$ and $N \iota y = 0$ has $p^k y = 0$, hence $y = 0$.

Step 3: the decomposition. Let $x \in \text{Cl}(L)_\ell$. Then $N(x) \in \text{Cl}(K)_\ell$, and we may put $y := u N(x)$ with u as in Step 2, so that $p^k y = N(x)$. Then $N(x - \iota y) = N(x) - N \iota y = N(x) - p^k y = 0$, i.e. $x - \iota y \in \ker N$; it lies in $\text{Cl}(L)_\ell$ as a difference of two elements of ℓ -power order. Thus $x = \iota y + (x - \iota y)$ exhibits $\text{Cl}(L)_\ell = \iota(\text{Cl}(K)_\ell) + \ker(N)_\ell$, and the sum is direct by Step 2.

Consequently

$$|\mathrm{Cl}(L)_\ell| = |\mathrm{Cl}(K)_\ell| \cdot |\ker(N)_\ell|.$$

Step 4: the valuation statement. For a finite abelian group G , $|G_\ell| = \ell^{v_\ell(|G|)}$, so $v_\ell(h_L)$ equals v_ℓ of $|\mathrm{Cl}(L)_\ell|$, and likewise for K . Taking v_ℓ of the displayed identity, $v_\ell(h_L) - v_\ell(h_K) = v_\ell(|\ker(N)_\ell|) \geq 0$, with equality iff $\ker(N)_\ell = 0$. Hence $v_\ell(h_L) > v_\ell(h_K)$ iff $\ker(N)_\ell \neq 0$.

Step 5: the Weber tower. Apply this with $K = B_{n-1}$, $L = B_n$, $p = 2$, $k = 1$: B_n/B_{n-1} is cyclic of degree 2 (Galois group $\langle \tau \rangle$). Then $v_\ell(h_n) \geq v_\ell(h_{n-1})$ for every odd ℓ , which is the odd part of the integrality $h_{n-1} \mid h_n$ ([9] Eq. (17): $k_n = h_n/h_{n-1} \in \mathbb{Z}$ and $k_n = [RE_n^+ : A_n]$), and

$$\ell \mid k_n \iff v_\ell(h_n) > v_\ell(h_{n-1}) \iff \ker(N)_\ell \neq 0 :$$

the odd primes dividing k_n are exactly the odd primes of the “new” part $\ker N$ of the class group. This is the sense in which k_n , and not h_n , is the natural object at a fixed layer; [9] Eq. (17) is the description of k_n by relative units that Theorem A uses. Nothing about $\ell = p = 2$ is claimed. $\square$

APPENDIX C. CHARACTERS OF G_n AND FROBENIUS ORBITS

$(\mathbb{Z}/q)^\times = \{\pm 1\} \times \langle 3 \rangle$ with 3 of order 2^n (from $3^{2^k} \equiv 1 + 2^{k+2} \pmod{2^{k+3}}$). Since $\sigma_{-a} = \sigma_a$ on B_n , $G_n = (\mathbb{Z}/q)^\times / \{\pm 1\}$ is cyclic of order 2^n , generated by the class of 3, and $\tau = \sigma^m$ is the class of $1 + q/2$. A character of G_n is an even character of $(\mathbb{Z}/q)^\times$; it has conductor exactly q iff it is nontrivial on $\{1, 1 + q/2\}$ iff $\chi(\tau) = -1$; then $\chi(3)^m = \chi(\tau) = -1$, so $\chi(3)$ is a primitive 2^n -th root of unity, and the m such characters give all m primitive roots. Lemma B is the orbit count of the same Frobenius on the roots of Φ_{2^n} over $\mathbb{F}_\ell$.

APPENDIX D. DETERMINANT NORMALISATION AND CONSTANTS

Lemma D.1 (0; group determinant). $D_n = 2^{-m/2} \prod_{\chi(\tau)=-1} |S(\chi)|$, where $S(\chi) = \sum_{g \in G_n} \chi(g) \log |g\varepsilon_n|$.

Proof. $D_n^2 = \det(H_n^\top H_n)$, and

$$(H_n^\top H_n)_{ik} = \sum_{j < 2m} \lambda_{i+j} \lambda_{k+j} = c(k-i), \quad c(t) = \sum_{j \bmod 2m} \lambda_j \lambda_{j+t},$$

an even function with $c(t+m) = -c(t)$. For $\omega^m = -1$ and $v = (\omega^i)_{i < m}$, $(H_n^\top H_n v)_i = \omega^i \mu(\omega)$ with $\mu(\omega) = \sum_{u < m} c(u) \omega^{-u} = \frac{1}{2} \sum_{u \bmod 2m} c(u) \omega^{-u} = \frac{1}{2} |\hat{\lambda}(\omega)|^2$, $\hat{\lambda}(\omega) = \sum_{j < 2m} \lambda_j \omega^j$, using the m -periodicity of $c(u) \omega^{-u}$. The m vectors v for the m roots of $x^m + 1$ are independent (Vandermonde), so $\det(H_n^\top H_n) = 2^{-m} \prod_\omega |\hat{\lambda}(\omega)|^2$; with $\omega = \chi(3)$, $\hat{\lambda}(\omega) = S(\chi)$. This gives Prop. D(i). An independent check of the normalisation $2^{-m/2}$: the eigenvector identity $H_n^\top H_n v = \frac{1}{2} |S(\chi)|^2 v$ was verified in ball arithmetic for $n = 2, \dots, 7$ with residual $< 10^{-143}$, and $\sqrt{\det H_n^\top H_n}$ agrees with $2^{-m/2} \prod |S(\chi)|$ to the precision of the determinant ball (`sage/r11_propD_audit.log`). $\square$

Lemma D.2 (1; a cyclotomic-unit expression). $\varepsilon_n = (1 - \zeta^3)^2(1 - \zeta^2)/((1 - \zeta)^2(1 - \zeta^6))$ with $\zeta = e^{2\pi i/q}$, $q = 2^{n+2}$.

Proof. With $\zeta = e^{2\pi i/q}$ and $X = \zeta + \zeta^{-1}$: $X + 1 = \zeta^{-1}(\zeta^3 - 1)/(\zeta - 1)$, $X - 1 = \zeta^{-1}(\zeta^3 + 1)/(\zeta + 1)$, hence $\varepsilon_n = (1 - \zeta^3)^2(1 - \zeta^2)/((1 - \zeta)^2(1 - \zeta^6))$; all factors are nonzero for $q \geq 16$. Consequently $\log |\sigma_a \varepsilon_n| = 2 \log |1 - \zeta^{3a}| - 2 \log |1 - \zeta^a| + \log |1 - \zeta^{2a}| - \log |1 - \zeta^{6a}|$ for $a \in (\mathbb{Z}/q)^\times$. $\square$

Lemma D.3 (2; imprimitive terms vanish). For χ of conductor exactly q and $c \in \{2, 6\}$: $\sum_{a \in (\mathbb{Z}/q)^\times} \chi(a) \log |1 - \zeta^{ca}| = 0$.

Proof. For χ of conductor exactly q and $c \in \{2, 6\}$, $\sum_{a \in (\mathbb{Z}/q)^\times} \chi(a) \log |1 - \zeta^{ca}| = 0$: ζ^{ca} depends only on $a \bmod q/2$, so the summand is constant on cosets of $U = \{1, 1 + q/2\}$, and $\sum_{u \in U} \chi(u) = 1 + \chi(1 + q/2) = 0$. $\square$

Lemma D.4 (3; Washington's formula in the form used). For χ even and primitive of conductor $f > 1$: $|\sum_{a \in (\mathbb{Z}/f)^\times} \chi(a) \log |1 - \zeta_f^a|| = \sqrt{f} |L(1, \chi)|$.

Proof. For χ even and primitive of conductor $f > 1$ ([20], Chapter 4), and with $|\tau(\chi)| = \sqrt{f}$,

$$L(1, \chi) = -\frac{\tau(\chi)}{f} \sum_{a=1}^f \bar{\chi}(a) \log |1 - \zeta_f^a|.$$

Applying this to $\bar{\chi}$ and using $|L(1, \bar{\chi})| = |L(1, \chi)|$:

$$\left| \sum_{a \in (\mathbb{Z}/f)^\times} \chi(a) \log |1 - \zeta_f^a| \right| = \sqrt{f} |L(1, \chi)|.$$

The conjugation convention does not affect the absolute value. $\square$

Lemma D.5 (4; $\prod |1 - \chi(3)| = 2$ for every n). For every $n \geq 2$: $\prod_{\chi(\tau)=-1} |1 - \chi(3)| = 2$.

Proof. By Appendix C (Frobenius orbits), as χ runs over the characters of G_n with $\chi(\tau) = -1$, i.e. the even characters of conductor exactly $q = 2^{n+2}$, the value $\chi(3)$ runs exactly once over the primitive 2^n -th roots of unity ω : indeed χ is determined by $\chi(3)$ because 3 generates G_n , and $\chi(3)^m = \chi(\tau) = -1$ with $m = 2^{n-1}$ says that $\chi(3)$ is a root of $x^m + 1 = \Phi_{2^n}(x)$, all of whose m roots are primitive 2^n -th roots of unity. Hence $\prod_{\chi} (1 - \chi(3)) = \prod_{\omega \text{ primitive}} (1 - \omega) = \Phi_{2^n}(1) = 1^m + 1 = 2$, and taking absolute values, $\prod_{\chi} |1 - \chi(3)| = 2$ for every $n \geq 2$. (The symbolic check in $\mathbb{Q}(\zeta_{2^n})$ for $n \leq 12$ in `sage/r11_propD_audit.log` is a numerical confirmation of this identity, not its proof.) $\square$

Proof of Prop. 1.4. (i) is Lemma D.1. (ii): Each $g \in G_n$ has two lifts $\pm a$ with the same $\log |\sigma_a \varepsilon_n|$, so $S(\chi) = \frac{1}{2} \sum_{a \in (\mathbb{Z}/q)^\times} \chi(a) \log |\sigma_a \varepsilon_n|$ (this half-sum normalisation was checked independently in ball arithmetic, $n = 2, \dots, 7$, residual $< 10^{-143}$). Insert Lemma D.2 and drop the $c = 2, 6$ terms by Lemma D.3: $S(\chi) = \sum_a \chi(a) \log |1 - \zeta^{3a}| - \sum_a \chi(a) \log |1 - \zeta^a| = (\bar{\chi}(3) -$

1) $\sum_a \chi(a) \log |1 - \zeta^a|$ by the substitution $b = 3a$. Lemma D.4 gives $|S(\chi)| = |1 - \chi(3)| \sqrt{q} |L(1, \chi)|$. Then Lemma D.1 with $\prod_\chi |1 - \chi(3)| = 2$ (Lemma D.5, every $n \geq 2$) and $q^{m/2} = 2^{(n+2)m/2}$: $D_n = 2^{-m/2+(n+2)m/2+1} \prod_\chi |L(1, \chi)| = 2^{m(n+1)/2+1} \prod_\chi |L(1, \chi)|$. $\square$

Proof of Cor. 1.5. $L_n^m = 2^{nm/2} \log(2 + \sqrt{5})^m$, and $2^{m(n+1)/2+1} / 2^{nm/2} = 2^{m/2+1}$, $(2/\pi)^{m/2} 2^{m/2} = (4/\pi)^{m/2}$. $\square$

Proof of Cor. 1.6. Let $B > 1$, $\ell \geq B$ and $d = d_n(\ell) \geq f_n(B)$, where $f_n(B) = \max\{1, 1 + \lfloor \log C_n / \log B \rfloor\}$ and $C_n > 0$ (Prop. D: $D_n > 0$).

The integer floor. Write $x = \log C_n / \log B \in \mathbb{R}$ (the sign of x is irrelevant; for $C_n < 1$ it is negative). By definition of the floor, $\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1$, so $f_n(B) \geq \lfloor x \rfloor + 1 > x$; this is the only property of $f_n(B)$ used, and it is why the “+1” is in the definition: $\lfloor x \rfloor + 1$ is the least integer strictly larger than x . The outer $\max\{1, \cdot\}$ only matters when $C_n < 1$ (then $x < 0$ and $\lfloor x \rfloor + 1 \leq 0$); it keeps $f_n(B) \geq 1 = \min_\ell d_n(\ell)$, so that the hypothesis $d \geq f_n(B)$ is never vacuous, and it does not affect the inequality $f_n(B) > x$. (Without the max the statement would still be true, because $d \geq 1$ for every odd ℓ ; the max removes the case distinction.)

From the floor to the constant. Since $B > 1$, $\log B > 0$ and $x \mapsto B^x = \exp(x \log B)$ is strictly increasing on $\mathbb{R}$. Hence $B^{f_n(B)} > B^x = \exp(\log C_n) = C_n$. If $d \geq f_n(B)$ then $B^d \geq B^{f_n(B)}$ (monotonicity again, $B > 1$), so $B^d > C_n$. If instead $d < f_n(B)$ is excluded by hypothesis, nothing else is needed: we only use $d \geq f_n(B)$ and $d \geq 1$.

From B to ℓ . $\ell \geq B > 1$ and $d \geq 1$ give $\ell^d \geq B^d$ (monotonicity of $y \mapsto y^d$ on $y > 0$ for a positive integer exponent). Therefore $\ell^{d_n(\ell)} = \ell^d \geq B^d > C_n$, which is the hypothesis $\ell^{d_n(\ell)} > C_n$ of Theorem 1.3, and Theorem 1.3 gives $\ell \nmid k_n$.

Example (the numbers of Cor. 7). At $n = 7$ with $B = 1314283897427173$, the certified integer comparisons give $B < C_7 < B^2$ (the second is Lemma D.7), so $1 < \log C_7 / \log B < 2$ and $f_7(B) = 1 + 1 = 2$; every prime $\ell \geq B$ with $d_7(\ell) \geq 2$, i.e. $\ell \not\equiv 1 \pmod{128}$, satisfies $\ell^2 \geq B^2 > C_7$ (the integer form of this comparison is Lemma D.7). With $B = 1727342163036353095979941756930$ the certified comparison gives $1 < C_7 < B$, so $0 < \log C_7 / \log B < 1$ and $f_7(B) = 1$, and every prime $\ell \geq B$ is covered whatever its order. Both thresholds are the frozen integers of Cor. 1.9.

Lean. `TheoremAInputs.theoremA_threshold` takes the two comparisons $C < B^d$ and $B^d \leq \ell^d$ as hypotheses and feeds them to the core `theoremA_of_inputs`; the floor computation above is the natural-language justification of those two hypotheses and is not itself formalised (it is ordinary real-number monotonicity). $\square$

Proof of Cor. 1.7. By Corollary 1.5,

$$C_n = 2(4/\pi)^{m/2} \Gamma(2 + m/2) \prod_{\chi} |L(1, \chi)| / \log(2 + \sqrt{5})^m,$$

the product over the m even Dirichlet characters of conductor exactly $q = 2^{n+2}$ (Proposition 1.4). Each such χ is primitive and even, $2 \mid q$ and $q \geq 16 \geq 4$, so [15] Corollary 1 with $h = 1$ and $k = 2$ gives $0 \leq |L(1, \chi)| \leq U(q)$. A product of m real numbers in $[0, U(q)]$ is at most $U(q)^m$; hence

$$C_n \leq 2(4/\pi)^{m/2} \Gamma(2 + m/2) U(q)^m / \log(2 + \sqrt{5})^m = \hat{C}_n.$$

If $\ell^{d_n(\ell)} > \hat{C}_n$ then $\ell^{d_n(\ell)} > C_n$, and Theorem 1.3 gives $\ell \nmid k_n$. The discrete part (a product of m factors in $[0, U]$ is $\leq U^m$, and $C \leq \hat{C} < \ell^d$ feeds $C < \ell^d$ into the discrete core `TheoremAInputs.theoremA_of_inputs`) is the Lean theorems `WeberHatC.prod_le_pow_card_of_le`, `WeberHatC.C_le_hatC` and `WeberHatC.theoremA_of_hat`; the analytic input $|L(1, \chi)| \leq U(q)$ is a literature hypothesis and is not formalised. $\square$

Proof of Cor. 1.8. Since $n \geq 2$, $m/2 = 2^{n-2}$ is an integer and $\Gamma(2 + m/2) = (1 + m/2)! = k!$ with $k = 1 + m/2 \geq 2$ (for a positive integer k , $\Gamma(k+1) = k!$). Robbins' form of Stirling's formula ([16]) gives $k! \leq \sqrt{2\pi k} (k/e)^k e^{1/(12k)}$, i.e. $\log k! \leq \frac{1}{2} \log(2\pi k) + k(\log k - 1) + \frac{1}{12k}$. Taking logarithms in the definition of $\hat{C}_n$ (all factors are positive: $U(q) > 0$, $\log(2 + \sqrt{5}) > 0$),

$$\begin{aligned} \log \hat{C}_n &= \log 2 + \frac{m}{2} \log \frac{4}{\pi} + \log k! + m \log \frac{U(q)}{\log(2 + \sqrt{5})} \\ &\leq \log 2 + \frac{m}{2} \log \frac{4}{\pi} + \frac{1}{2} \log(2\pi k) + k(\log k - 1) + \frac{1}{12k} \\ &\quad + m \log \frac{U(q)}{\log(2 + \sqrt{5})} = T_n. \end{aligned}$$

If $d_n(\ell) \log \ell > T_n$ then $\ell^{d_n(\ell)} = e^{d_n(\ell) \log \ell} > e^{T_n} \geq \hat{C}_n$, and Corollary 1.7 gives $\ell \nmid k_n$.

Check at $n = 7$. $m = 64$, $k = 33$, $q = 512$, $U(512) = 2.2431\dots$: $T_7 = \log 2 + 32 \log(4/\pi) + \frac{1}{2} \log(66\pi) + 33(\log 33 - 1) + \frac{1}{396} + 64 \log(2.24312/1.44364) = 0.69315 + 7.73006 + 2.66719 + 82.38475 + 0.00253 + 28.20496 = 121.68264$, while $\log \hat{C}_7 = 121.68264$ ($\hat{C}_7 = 7.016 \times 10^{52}$, `sage/r16_hatCn/r16_hatCn_pilot.log`); $T_7 - \log \hat{C}_7 = 7.7 \times 10^{-8}$, i.e. $1/(12k) - r_{33}$ with r_{33} the Robbins remainder ($1/397 < r_{33} < 1/396$). Every constant is displayed in T_n ; the remark that $T_n = \frac{m}{2} \log m + m \log(n+2) + O(m)$ (from $\log k! = k \log k - k + O(\log k)$ and $U(2^{n+2}) = \frac{(n+2) \log 2}{4} + O(1)$) is descriptive and is not part of the statement. $\square$

Lemma D.6 (C7; rigorous enclosure). $C_7 \in [C_7^-, C_7^+]$, where both endpoints are recorded to 160 significant digits, rounded outward, in `certificates/constants/Cn_interval_r14.json` (digamma route, 500-bit ball arithmetic, `sage/r14_cn_interval.sage/.log`;

$C_7^+ - C_7^- < 5 \times 10^{-115}$; the two endpoints share their first 145 digits). The displayed value $C_7 = 1.7273421630363529579743237623519834054 \dots \times 10^{30}$ is the truncation of that common prefix to 38 digits (truncation error $< 10^{-7}$; the width of the enclosure is a property of the certificate, not of the displayed digits). The enclosure is consistent with the interval determinant of `certificates/blichfeldt/r6_blichfeldt_cert.log`, the PARI Gauss-sum route and the Magma evaluation (Section 4); in particular $C_7^+ < T_1 = 1727342163036353095979941756929$ (asserted inside the certificate against the frozen integer of Lemma D.7).

Lemma D.7 (C7; integer comparison). *For integers ℓ :*

$$1314283897427172 < \ell \implies 1727342163036353095979941756929 < \ell^2.$$

Note. Lean: `c7_deg2_instance`, `deg1_threshold_valid`, `deg2_threshold_valid`.

Proof of Cor. 1.9. Take $n = 7$, $m = 64$, $k_7 = h_7/h_6$. By Lemma 2.7, $d_7(\ell) = 1$ for $\ell \equiv 1 \pmod{128}$ and $d_7(\ell) = 2$ for $\ell \equiv 63, 65, 127 \pmod{128}$. Lemma D.6 encloses C_7 in a rigorous ball whose upper end is below the integer $T_1 = 1727342163036353095979941756929$, and $T_2 = 1314283897427172$ satisfies $T_1 < \ell^2$ whenever $T_2 < \ell$ (Lemma D.7, kernel-checked; the comparison $C_7 < T_1$ and T_2^2 against the ball are verifier step 04). Hence $\ell > T_1$ gives $\ell^1 > C_7$, and $\ell > T_2$ with $d_7(\ell) = 2$ gives $\ell^2 > T_1 > C_7$; in both cases Theorem 1.3 yields $\ell \nmid k_7$. $\square$

Citation note. The edition used for [20] is the second (GTM 83, 1997); the two inputs taken from it are the formula for $L(1, \chi)$ with χ even and primitive and the Gauss-sum modulus $|\tau(\chi)| = \sqrt{f}$, both from Chapter 4, and the discriminant and integral basis of $\mathbb{Q}(\zeta_{p^r})$ from Chapter 2; they are cited at chapter level. The $L(1, \chi)$ identity is standard and is also verified numerically (Table 5, three routes).

APPENDIX E. THE SCALING THEOREM

Lemma E.1 (S1; group lemma). *Let R be an abelian group, $L \leq R$ of finite index, and k an integer coprime to $[R : L]$. Then $L \cap kR = kL$. (Lean: `smul_mem_iff_of_coprime`, `coprime_smul_injective`.)*

Proof. Let R be an abelian group (written additively), $L \leq R$ a subgroup of finite index $N = [R : L]$, and $k \geq 1$ an integer with $\gcd(k, N) = 1$. We show $L \cap kR = kL$, where $kR = \{kr : r \in R\}$ and $kL = \{ky : y \in L\}$.

$kL \subseteq L \cap kR$. If $x = ky$ with $y \in L$, then $x \in kR$ by definition, and $x \in L$ because a subgroup is closed under integer multiples.

$L \cap kR \subseteq kL$. Let $x \in L \cap kR$, say $x = kr$ with $r \in R$, and let $\pi : R \rightarrow R/L$ be the projection. Since π is a homomorphism, $\pi(x) = k\pi(r)$, and $\pi(x) = 0$ because $x \in L$; hence $k\pi(r) = 0$ in the finite group R/L of order N . Multiplication by k is injective on R/L : if $k\bar{y} = 0$, choose integers a, b with $ak + bN = 1$ (possible since $\gcd(k, N) = 1$); then $\bar{y} =$

$(ak + bN)\bar{y} = a(k\bar{y}) + b(N\bar{y}) = 0$, because $N\bar{y} = 0$ by Lagrange's theorem ($\bar{y}$ has order dividing $|R/L| = N$). (This is `coprime_smul_injective`: on a finite abelian group whose cardinality is coprime to k , $y \mapsto ky$ is injective.) Applied to $\bar{y} = \pi(r)$ it gives $\pi(r) = 0$, i.e. $r \in L$, and therefore $x = kr \in kL$.

The two inclusions together are the equivalence $kr \in L \iff r \in L$ for every $r \in R$ (`smul_mem_iff_of_coprime`, whose hypotheses are exactly: R an additive commutative group, L a subgroup with R/L finite, and $\gcd(k, |R/L|) = 1$). In the application (Theorem 1.10) $R = R_n$, $L = L_f$ (notation of the setting), $[R_n : L_f]$ is a power of the odd prime ℓ and $k = 2^{t-1}$, so $\gcd(k, [R_n : L_f]) = 1$ and $L_f \cap 2^{t-1}R_n = 2^{t-1}L_f$. $\square$

R_n/L_f is an $\mathbb{F}_\ell$ -vector space (Lemma C), so $[R_n : L_f]$ is a power of the odd prime ℓ , and Lemma S1 applies with $k = 2^{t-1}$: $L_f \cap 2^{t-1}R_n = 2^{t-1}L_f$; likewise $\ell R_n \cap 2^{t-1}R_n = 2^{t-1}\ell R_n$.

Lemma E.2 (depth- t height floor). *For $t \geq 1$ and $u \in RE_n^+ \setminus \{\pm 1\}$ with $u \equiv 1 \pmod{2^t \mathcal{O}_n}$: $\text{ht}(u) \geq L_{n,t} = \sqrt{2^n} \operatorname{arcsinh}(2^t)$.*

Proof. Let $t \geq 1$ and $u \in RE_n^+ \setminus \{\pm 1\}$ with $u \equiv 1 \pmod{2^t \mathcal{O}_n}$. By Lemma 2.5, $u \notin B_{n-1}$; since $B_n/\mathbb{Q}$ is cyclic of 2-power degree, its subfields are the B_k , so $\mathbb{Q}(u) = B_n$ and u has degree $N = 2^n$. From $u - 1 \in 2^t \mathcal{O}_n$ and $u + 1 = (u - 1) + 2 \in 2\mathcal{O}_n$ we get $u^2 - 1 \in 2^{t+1} \mathcal{O}_n$, so the ideal $\mathfrak{M} = 2^{t+1} \mathcal{O}_n$ contains $u^2 - 1$ and has absolute norm $C = 2^{(t+1)N}$, $C^{1/N} = 2^{t+1}$. Schinzel's inequality in the form quoted as [14] Thm. 5.1 ("Let ε be a totally real unit different from ± 1 and $\mathfrak{M}$ an ideal of $\mathbb{Q}(\varepsilon)$ containing $\varepsilon^2 - 1$; then $M(\varepsilon) \geq ((C^{1/d} + \sqrt{C^{2/d} + 4})/2)^{d/2}$, C the absolute norm of $\mathfrak{M}$, d the degree") gives, with $d = N$, $M(u) \geq ((2^{t+1} + \sqrt{4^{t+1} + 4})/2)^{N/2} = (2^t + \sqrt{4^t + 1})^{N/2} = \exp(\frac{N}{2} \operatorname{arcsinh}(2^t))$, using $\operatorname{arcsinh} x = \log(x + \sqrt{x^2 + 1})$. Finally, for a unit the conjugates satisfy $\sum_i \log |u_i| = 0$, so $\log M(u) = \sum_{|u_i| > 1} \log |u_i| = \frac{1}{2} \sum_i |\log |u_i|| \leq \frac{1}{2} \sqrt{N} \text{ht}(u)$ by Cauchy-Schwarz (this is [13] eq. (2.4), proof of their Lemma 2.3). Hence $\text{ht}(u) \geq \frac{2}{\sqrt{N}} \cdot \frac{N}{2} \operatorname{arcsinh}(2^t) = \sqrt{2^n} \operatorname{arcsinh}(2^t) = L_{n,t}$. For $t = 1$ this is $\sqrt{2^n} \log(2 + \sqrt{5}) = L_n$, because $\operatorname{arcsinh} 2 = \log(2 + \sqrt{5})$. (This floor is proved here for every $t \geq 1$, so Theorem 1.10 keeps only hypothesis (i), which remains open for $t \geq 5$.) $\square$

Lemma E.3 (odd-power depth transfer). *Let $t \geq 1$, ℓ an odd integer and $x, y \in \mathcal{O}_n^\times$. Then $x^\ell \equiv 1 \pmod{2^t \mathcal{O}_n}$ if and only if $x \equiv 1 \pmod{2^t \mathcal{O}_n}$; more generally $x^\ell \equiv y^\ell \pmod{2^t \mathcal{O}_n}$ if and only if $x \equiv y \pmod{2^t \mathcal{O}_n}$.*

Note. Lean, group core: `WeberOddTransfer.pow_odd_eq_one_iff_of_two_pow_card`, `pow_odd_iff_of_two_pow_card`, `isUnit_pow_odd_eq_one_iff`; the arithmetic of B_n in the proof is [M].

Proof. Write $N = 2^n = [B_n : \mathbb{Q}]$, $q = 2^{n+2}$, $\zeta = e^{2\pi i/q}$, $K = \mathbb{Q}(\zeta)$ and $\pi = 1 - \zeta$; $B_n = \mathbb{Q}(\zeta + \zeta^{-1}) \subset K$ with $[K : B_n] = 2$.

Step 1: 2 is totally ramified in B_n with residue field $\mathbb{F}_2$. For odd a with $0 < a < q$ the quotient $(1 - \zeta^a)/(1 - \zeta) = 1 + \zeta + \dots + \zeta^{a-1}$ lies in $\mathbb{Z}[\zeta]$, and if $ab \equiv 1 \pmod{q}$ then $(1 - \zeta)/(1 - \zeta^a) = (1 - \zeta^{ab})/(1 - \zeta^a) = 1 + \zeta^a + \dots + \zeta^{a(b-1)} \in \mathbb{Z}[\zeta]$; so $1 - \zeta^a = \pi \eta_a$ with $\eta_a \in \mathbb{Z}[\zeta]^\times$. The q -th cyclotomic polynomial is $\Phi_q(X) = X^{q/2} + 1 = \prod_{a \text{ odd}} (X - \zeta^a)$, hence $\prod_{a \text{ odd}} (1 - \zeta^a) = \Phi_q(1) = 2$, i.e. $2 = \eta \pi^{\varphi(q)}$ with η a unit and $\varphi(q) = 2^{n+1} = [K : \mathbb{Q}]$. Since the ramification index of a prime above 2 is at most $[K : \mathbb{Q}]$, this forces $2\mathcal{O}_K = \mathfrak{P}^{[K:\mathbb{Q}]}$ with $\mathfrak{P} = \pi\mathcal{O}_K$ the unique prime of K above 2, and its residue degree is 1. Let $\mathfrak{p} = \mathfrak{P} \cap \mathcal{O}_n$. Every prime of $\mathcal{O}_n$ above 2 lies below $\mathfrak{P}$, so $\mathfrak{p}$ is the unique such prime, and ramification indices multiply in the tower $\mathbb{Q} \subset B_n \subset K$: $[K : \mathbb{Q}] = e(\mathfrak{P}/2) = e(\mathfrak{P}/\mathfrak{p})e(\mathfrak{p}/2) \leq [K : B_n][B_n : \mathbb{Q}] = [K : \mathbb{Q}]$, whence $e(\mathfrak{p}/2) = [B_n : \mathbb{Q}] = N$. Then $2\mathcal{O}_n = \mathfrak{p}^N$, the residue degree $f(\mathfrak{p}/2)$ is 1 (because $ef \leq N$), $\mathcal{O}_n/\mathfrak{p} = \mathbb{F}_2$, and $2^t\mathcal{O}_n = \mathfrak{p}^{Nt}$ for every $t \geq 1$.

Step 2: the unit group of $\mathcal{O}_n/2^t\mathcal{O}_n$ is a 2-group. Put $k = Nt \geq 1$. The ring $\bar{R} = \mathcal{O}_n/\mathfrak{p}^k$ is finite of order $N(\mathfrak{p}^k) = N(\mathfrak{p})^k = 2^k$ (multiplicativity of the absolute norm in the Dedekind domain $\mathcal{O}_n$), it is local with unique maximal ideal $\bar{\mathfrak{p}} = \mathfrak{p}/\mathfrak{p}^k$ (every ideal of $\bar{R}$ is $\bar{\mathfrak{p}}^i/\mathfrak{p}^k$, $0 \leq i \leq k$), and $|\bar{\mathfrak{p}}| = N(\mathfrak{p})^{k-1} = 2^{k-1}$. In a finite local ring the units are exactly the elements outside the maximal ideal, so $|\bar{R}^\times| = 2^k - 2^{k-1} = 2^{k-1}$.

Step 3: the odd power map on $\bar{R}^\times$ is a bijection. **[F]:WeberOddTransfer.** `pow_odd_eq_one_iff_of_two_pow_card` (a group G with $|G| = 2^{k'}$ and ℓ odd satisfy $g^\ell = 1 \iff g = 1$), `pow_odd_eq_iff_of_two_pow_card` ($g^\ell = h^\ell \iff g = h$) and the ring form `isUnit_pow_odd_eq_one_iff`; the proof is Mathlib's `powCoprime`: when $\gcd(|G|, \ell) = 1$ the map $g \mapsto g^\ell$ is a bijection of G with inverse $g \mapsto g^b$, $b\ell \equiv 1 \pmod{|G|}$. Here $|\bar{R}^\times| = 2^{k-1}$ and ℓ is odd.

Step 4: transfer. Let $x \in \mathcal{O}_n^\times$ and let $\bar{x}$ be its image in $\bar{R} = \mathcal{O}_n/2^t\mathcal{O}_n$; $\bar{x} \in \bar{R}^\times$ because x is a unit, and for any $y \in \mathcal{O}_n$ one has $y \equiv 1 \pmod{2^t\mathcal{O}_n}$ iff $\bar{y} = 1$. Hence $x^\ell \equiv 1 \pmod{2^t\mathcal{O}_n} \iff \bar{x}^\ell = 1 \iff \bar{x} = 1 \iff x \equiv 1 \pmod{2^t\mathcal{O}_n}$, the middle equivalence being Step 3. For units x, y the same argument applied to $xy^{-1} \in \mathcal{O}_n^\times$ gives $x^\ell \equiv y^\ell \iff x \equiv y \pmod{2^t\mathcal{O}_n}$.

Remark (valuation form of the same fact). Since $\mathcal{O}_n/\mathfrak{p} = \mathbb{F}_2$, every unit satisfies $x \equiv 1 \pmod{\mathfrak{p}}$, so $1 + x + \dots + x^{\ell-1} \equiv \ell \equiv 1 \pmod{\mathfrak{p}}$ is a $\mathfrak{p}$ -adic unit and $v_{\mathfrak{p}}(x^\ell - 1) = v_{\mathfrak{p}}(x - 1) + v_{\mathfrak{p}}(1 + x + \dots + x^{\ell-1}) = v_{\mathfrak{p}}(x - 1)$; the two congruences of the lemma are the two conditions $v_{\mathfrak{p}}(\cdot) \geq Nt$ on these equal quantities. This second proof is recorded for the reader; the proof above is the one whose discrete core is kernel-checked. $\square$

Proof of Theorem 1.10. The lattice half. R_n/L_f is an $\mathbb{F}_\ell$ -vector space (Lemma 2.8), so $[R_n : L_f]$ is a power of the odd prime ℓ and Lemma S1 (the group lemma) applies with $k = 2^{t-1}$: $L_f \cap 2^{t-1}R_n = 2^{t-1}L_f$; likewise $\ell R_n \cap 2^{t-1}R_n = 2^{t-1}\ell R_n$. Under (i), $L_f \cap J_t = L_f \cap 2^{t-1}R_n = 2^{t-1}L_f$, hence $\Lambda_{f,t} = \ell^{-1}H_n(2^{t-1}L_f) = 2^{t-1}\Lambda_f$, and the vectors of $\Lambda_{f,t}$ coming from

$a \notin \ell R_n$ are exactly the 2^{t-1} -dilates of those of Λ_f coming from $a \notin \ell R_n$ (the same lemma for ℓR_n).

What the depth- t variant needs. The depth- t variant of Theorem A is the argument of Prop. F with Λ_{f_0} replaced by $\Lambda_{f_0,t}$ and the floor L_n of Lemma A⁺ replaced by the depth- t floor $L_{n,t}$ of Lemma E.2. We make explicit the unit to which the floor is applied, in both branches of Prop. F. Suppose $\ell \mid k_n$, let f_0 be the saturation component of Lemma 2.3, and let $v \in \Lambda_{f_0,t}$ be a nonzero vector; write $v = \ell^{-1}H_n a$ with $a \in L_{f_0} \cap J_t$, $a \neq 0$. Since $a \in J_t$, $u_a \equiv 1 \pmod{2^t \mathcal{O}_n}$. (Non-base branch, $a \notin \ell R_n$.) By Lemma 2.3, $r_a \in RE_n^+$ and v is the log vector of r_a , so $\text{ht}(r_a) = |v|$; $r_a \neq \pm 1$ because $v \neq 0$. The unit r_a is a formal ℓ -th root: $r_a^\ell = u_a \equiv 1 \pmod{2^t \mathcal{O}_n}$. Lemma E.3 (odd-power depth transfer, $x = r_a$) gives $r_a \equiv 1 \pmod{2^t \mathcal{O}_n}$, and Lemma E.2 gives $\text{ht}(r_a) \geq L_{n,t}$. (Base branch, $a = \ell b$, $b \in R_n \setminus \{0\}$.) By Lemma 2.4, $v = H_n b$ is the log vector of $u_b \in A_n \subseteq RE_n^+$ and $u_b \neq \pm 1$; here $u_b^\ell = u_{\ell b} = u_a \equiv 1 \pmod{2^t \mathcal{O}_n}$, so Lemma E.3 ($x = u_b$) gives $u_b \equiv 1 \pmod{2^t \mathcal{O}_n}$ and Lemma E.2 gives $\text{ht}(u_b) \geq L_{n,t}$. In both branches the unit carrying v has height $|v| \geq L_{n,t}$. Therefore the depth- t variant reaches its contradiction exactly when Blichfeldt supplies a nonzero $v \in \Lambda_{f_0,t}$ with $|v| < L_{n,t}$; without Lemma E.3 the passage from the depth of $u_a = r_a^\ell$ (the quantity filtered by J_t) to the depth of r_a (the unit whose height is bounded) would be unjustified.

Comparison. By the lattice half, a nonzero $v \in \Lambda_{f_0,t}$ with $|v| < L_{n,t}$ is $v = 2^{t-1}v'$ with a nonzero $v' \in \Lambda_{f_0}$ and $|v'| < L_{n,t}/2^{t-1}$, whereas Theorem A (Prop. F) needs a nonzero $v' \in \Lambda_{f_0}$ with $|v'| < L_n$. Now $L_{n,t}/2^{t-1} < L_n$ iff $\text{arcsinh}(2^t) < 2^{t-1} \log(2 + \sqrt{5})$ iff $2^t + \sqrt{4^t + 1} < (2 + \sqrt{5})^{2^{t-1}}$, and the last inequality holds for every $t \geq 2$ (Lemma S-analytic, [F]: `ineq_t_general`: $2^t + \sqrt{4^t + 1} < 2^{t+1} + 1 \leq 4^t \leq 4^{2^{t-1}} < (2 + \sqrt{5})^{2^{t-1}}$). Hence every vector that satisfies the depth- t requirement satisfies the requirement of Theorem A, and the requirement of the depth- t variant is strictly stronger: the direct filtered-lattice sufficient condition at depth t is stricter than that of Theorem A, i.e. the resulting direct filtered-lattice criterion is weaker. This is the only sense of “weaker” used in this paper; it concerns the direct application of the same Blichfeldt argument to the deeper congruence sublattice $\Lambda_{f,t}$ and says nothing about other 2-adic refinements. Nothing is asserted about the truth of hypothesis (i) beyond the certified cases of Cor. S1. $\square$

Proof of Cor. 1.11. For $t = 2, 3, 4$ at $n = 7$ the equality $J_t = 2^{t-1}R_n$ was established by exact rank computations of the 2-adic filtration of A_n , constructed independently in Sage (`sage/r8_twoadic_depth_exact.sage`) and in Magma (`sage/r8_twoadic_rank_magma.m`), and replayed by a standalone Python rank checker that recomputes the rank of the saved bit matrices (`scripts/verify_twoadic_rank.py`, certificates in `certificates/twoadic_rank/`; log `sage/r9_scaling_no_go.log`); the Python step is a replay of stored certificates, not a third construction. Lemma E.3 and

Lemma E.2 hold for every $t \geq 1$ (the latter relative to its literature inputs, Schinzel and MO16 eq. (2.4)), so Theorem 1.10 applies at $t = 2, 3, 4$ and the direct filtered-lattice routes at depths 4, 8 and 16 are strictly weaker than Theorem A. $\square$

APPENDIX F. ARTIFACT INDEX

File paths are given here once; the body of the paper refers to them by name. Every path is relative to the repository root; the SHA-256 of every file is in `MANIFEST_SHA256.txt`; the round-by-round correspondence is `CORRESPONDENCE.csv` (generated from `docs/CLAIMS_1.1.0.yaml`) and `docs/BLEUPRINT_MAP_1.1.0.md`.

Lean (kernel-checked, *std-3*). `lean/WeberR7.lean` (Lemmas A, C, coset absorption); `lean/WeberExternalResults.lean` (Theorem A core `TheoremAInputs.theoremA_of_inputs`, Cor. T `theoremA_threshold`, $n = 7$ comparison `c7_deg2_instance`); `lean/WeberCertChainDirect.lean` and `lean/WeberCertFloor.lean` (Theorem 1.12 core and the threshold $\bar{T}_n$); `lean/WeberScalingNoGo.lean` and `lean/WeberScalingS0.lean` (Theorem S0: group lemma, `ineq_t_general`); `lean/WeberOddTransfer.lean` (Lemma E.3, group core); `lean/WeberSH.lean` (Theorem 1.1: the typed bridge `WeberSH.sat_root_log`, the base branch `WeberSH.base_branch`, `WeberSH.theoremSH`); `lean/WeberHatC.lean` (Cor. 1.7: `WeberHatC.C_le_hatC`, `WeberHatC.theoremA_of_hat`); `lean/WeberLemmaB.lean` (Lemma 2.7 in full: `WeberLemmaB.factor_degree`); `lean/WeberRoots.lean` (uniqueness of the real root and the representative change of Lemma 2.3); `lean/WeberP3.lean` (Theorem 6.5: the discrete core with the $p = 3$ names; its Cauchy–Schwarz lemma served an earlier floor and is no longer load-bearing); `lean/WeberP3Rel.lean` (Lemma 6.1: order of $1 + 3^n$, action of τ , telescoping, periodicity, odd-power transfer; Theorem 6.3: the coset-sum vanishing of the Fourier coefficients at $3 \mid k$, the shift algebra of the twisted sums on $\mathbb{Z}/N$ that reduces $\hat{\lambda}_k \neq 0$ to $\hat{\Gamma}_k \neq 0$ for $3 \nmid k$, the primitive-root polynomial step and $\varphi(3^r) = 2 \cdot 3^{r-1}$). *Coverage of Theorem A*: kernel-checked are Lemmas A, B, C, the group core of Lemma E (coset absorption, real roots, the bridge $u^\ell = u_a \Rightarrow \ell H(u) = H(u_a)$), the base branch, Theorem SH’s discrete core and its two-branch conclusion, Cor. T and the $n = 7$ comparison; the remaining manuscript steps are Lemma D (covolume via Dirichlet’s unit theorem), the instantiation of Theorem SH with the Weber data, and Prop. D. The label of Theorem A stays “L-relative, F-core”; “kernel-only” is not claimed. Compilation logs with axiom census: `lean/compile*.log`; `lean/README_lean.md`.

Certificates. C_n enclosures: `certificates/constants/Cn_interval_r14.json` (both endpoints to 160 digits, outward; produced by `sage/r14_cn_interval.sage`); cross-CAS routes: `certificates/constants/C7_cross_cas_r14.json` (`sage/r14_cross_cas_audit.sage`); $r6$ interval determinant: `certificates/blichfeldt/r6_blichfeldt_cert.log`;

family: `certificates/family/KY1000/witness_n7_l*.txt` (1000 files),
 ledger `sage/family_ky1000_r11_clean/VERIFY_LEDGER.txt`; negative
 controls: `sage/negctl_r13/negctl_r13.log`, ledger `certificates/
 negctl/negctl_ledger_r14.json`; 2-adic rank certificates (Cor. S1):
`certificates/twoadic_rank/`; flagship prime: `certificates/flagship/`.
 $p = 3$ covolumes and constants (Table 8, Cor. 6.6): **[P3-CERT]** =
`certificates/p3/D3_cert_r19.json` (format v3: every certified quantity
 as EXACT dyadic endpoints, the outward hull of the two independent en-
 closures widened to twice its half-width, with both enclosures and the bare
 hull kept per field; 4000 bits); **[P3-GEN]** = `tools/gen_p3_cert_r19.py`
 from the producer balls `sage/r19_trackB/p3_covol_balls_r19.json`
 (`sage/r19_trackB/p3_covol_balls_r19.sage`) and the read-only re-
 computation `sage/r19_trackB/p3_readonly_recomputed_r19.json`;
[P3-REPLAY] = `scripts/verify_p3_readonly.sage` (rebuilds
 everything from the definitions; requires containment of its recom-
 putation in every certified interval, 120/120; twelve planted cer-
 tificates rejected, ledger `docs/P3_NEGCTL_LEDGER_R19.md`); **[P3-
 COVER]** = `tools/check_p3_containment.py`; **[P3-CHECK]** =
`tools/check_p3_cert.py` (consistency and print strings).

Checkers and scripts. Family verifier `scripts/family_verify.sage`
 (frozen; SHA-256 0f38bb0d...); target-list verifier `scripts/verify_
 ky1000_target.py`; 2-adic rank replay `scripts/verify_twoadic_rank.
 py`; generator (untrusted) `scripts/family_gen.sage`; portable re-
 play `scripts/verify_all_portable.sh` with the gates `tools/check_
 constant_sync.py`, `tools/negctl_tools_r14.py`, `tools/gen_cn_certs.
 py`, `tools/gen_negctl_ledger.py`, `tools/verifier_profile_tests.sh`,
`tools/check_release_metadata.py` (current-facing pointers and for-
 bidden notation), `tools/check_statement_sync.py` (the statement
 of Theorem 6.5 identical in this paper and in the Blueprint; the
 conditions of the claims ledger present in both); **[NOVELTY]** =
`docs/NOVELTY_MATRIX.md` (generated by `tools/gen_novelty_matrix.py`);
 CI `.github/workflows/verify.yml`, `Dockerfile`.

Documents. The Blueprint (statements, dependency graph and the same
 single-source proofs as this paper) and the repository’s records of frozen
 statements, trust legend, release status and errata accompany the code; the
 repository README lists them.

REFERENCES

- [1] H. F. Blichfeldt, A new principle in the geometry of numbers, with some applications,
 Trans. Amer. Math. Soc. 15 (1914), 227–235.
- [2] T. Fukuda, K. Komatsu, Weber’s class number problem in the cyclotomic $\mathbb{Z}_2$ -
 extension of $\mathbb{Q}$, III, Int. J. Number Theory 7 (2011), 1627–1635.
- [3] K. Horie, Ideal class groups of Iwasawa-theoretical abelian extensions over the rational
 field, J. London Math. Soc. (2) 66 (2002), 257–275.
- [4] K. Horie, The ideal class group of the basic $\mathbb{Z}_p$ -extension over an imaginary quadratic
 field, Tohoku Math. J. 57 (2005), 375–394.

- [5] K. Horie, Triviality in ideal class groups of Iwasawa-theoretical abelian number fields, *J. Math. Soc. Japan* 57 (2005), 827–857.
- [6] K. Horie, Primary components of the ideal class group of the $\mathbb{Z}_p$ -extension over $\mathbb{Q}$ for typical inert primes, *Proc. Japan Acad. Ser. A* 81 (2005), 40–43.
- [7] K. Horie, Certain primary components of the ideal class group of the $\mathbb{Z}_p$ -extension over the rationals, *Tohoku Math. J.* 59 (2007), 259–291.
- [8] K. Horie, M. Horie, The ideal class group of the $\mathbb{Z}_p$ -extension over the rationals, *Tohoku Math. J.* 61 (2009), 551–570.
- [9] T. Kashio, H. Yoshizaki, Minimal relative units of the cyclotomic $\mathbb{Z}_2$ -extension, *arXiv:2107.08587v3* (2022).
- [10] J. C. Miller, Class numbers of totally real fields and applications to the Weber class number problem, *Acta Arith.* 164 (2014), 381–397.
- [11] T. Morisawa, Mahler measure of the Horie unit and Weber’s class number problem in the cyclotomic $\mathbb{Z}_3$ -extension of $\mathbb{Q}$, *Acta Arith.* 153 (2012), 35–49.
- [12] T. Morisawa, R. Okazaki, Mahler measure and Weber’s class number problem in the cyclotomic $\mathbb{Z}_p$ -extension of $\mathbb{Q}$ for odd prime number p , *Tohoku Math. J.* 65 (2013), 253–272.
- [13] T. Morisawa, R. Okazaki, Height and Weber’s class number problem, *J. Théor. Nombres Bordeaux* 28 (2016), 811–828.
- [14] T. Morisawa, R. Okazaki, Filtrations of units of Viète field, *Int. J. Number Theory* 16 (2020), 1067–1079.
- [15] O. Ramaré, Approximate formulae for $L(1, \chi)$, II, *Acta Arith.* 112 (2004), 141–149.
- [16] H. Robbins, A remark on Stirling’s formula, *Amer. Math. Monthly* 62 (1955), 26–29.
- [17] A. Schinzel, On the product of the conjugates outside the unit circle of an algebraic number, *Acta Arith.* 24 (1973), 385–399.
- [18] L. C. Washington, Class numbers and $\mathbb{Z}_p$ -extensions, *Math. Ann.* 214 (1975), 177–193.
- [19] L. C. Washington, The non- p -part of the class number in a cyclotomic $\mathbb{Z}_p$ -extension, *Invent. Math.* 49 (1978), 87–97.
- [20] L. C. Washington, *Introduction to Cyclotomic Fields*, 2nd ed., GTM 83, Springer, 1997.

RESEARCH INSTITUTE OF CRIMINAL PSYCHIATRY / SEX OFFENDER MEDICAL CENTER,
TOKYO, JAPAN; AND DEPARTMENT OF NEUROPSYCHIATRY, KYOTO UNIVERSITY, KYOTO,
JAPAN. ORCID: 0009-0008-7122-522X

Email address: fukui@somec.org